Parity Statistics of Nested Floor Powers

Finite-Step Descent and Conditional Extensions for the Juggler Map

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Abstract

The Juggler map applies the integer part of the square root at even positive integers and of the three-halves power at odd positive integers. We prove that the starting values admitting a power-envelope descent certificate within five operations have natural density 7/8. More precisely, their count up to N is $7N/8 + O_\varepsilon(N^{127/128+\varepsilon})$ for every $\varepsilon > 0$. The four-step subfamily has density 13/16. The proof uses exact carry identities, centered Fourier expansions, and van der Corput differencing, with all floor exceptions and partition endpoints counted. The two five-letter classes *OOEOE* and *OOOEE* are treated through their distinct formal chains. For *OOOEE*, the full mixed phase is retained and its signed curvature is recomputed after each frequency center is frozen. The complete analytic argument is included in three appendices. Fair-share densities at every fixed depth would imply density-one finite certificates; that hypothesis, general decorated estimates, and short-interval localization remain open. No result asserts universal arrival at 1.

Keywords: Juggler map; nested floor powers; parity correlations; discrepancy; stopping time; finite-step descent.

1. Scope and relation to earlier work

For a positive integer n, define

$$J(n) = \begin{cases} \lfloor n^{1/2} \rfloor, & n \text{ even}, \\ \lfloor n^{3/2} \rfloor, & n \text{ odd}. \end{cases}$$

This is the Juggler map recorded in OEIS A094683 [1]. The conjecture that every positive orbit reaches 1 remains open. The question studied here is more limited: which distribution estimates for finite itineraries would imply that many starting values eventually fall below their initial value?

Single-floor parity estimates follow from classical exponential-sum methods [2, 3]. Composing floors creates an additional difficulty. For $m = \lfloor n^{3/2} \rfloor$ and $\theta = \{n^{3/2}\}$,

$$m^{3/2} = n^{9/4} - \tfrac{3}{2}\theta n^{3/4} + O(n^{-3/4}).$$

The coefficient of the fractional part grows with n. A theorem for the smooth phase $n^{9/4}$ therefore does not by itself estimate the nested phase. Nor does a parity estimate over consecutive starting values automatically apply to a sparse set of orbit images.

The main result is the five-step certificate count in Theorem 5.4. The proof distinguishes finite word classification from the analytic estimates needed to count those words. Sections 4.2–4.3 establish the four-step estimate and the *OOEOE* split. Section 4.4 and Appendices A–C establish the remaining *OOOEE* mixed-mode bound. All signs, zero coordinates, Fourier coefficient weights, and interval boundaries are included.

The single-floor estimate and the conditional all-depth counting argument use standard methods. The latter is in the spirit of stopping-time arguments for the Collatz map, such as Terras [4]; no theorem about Collatz is transferred to the Juggler map. The general 95/96 kernel target and arbitrary short-interval extensions from earlier drafts remain unproved. The weaker 127/128 exponent proved here suffices for the five-step count.

This preprint supersedes the earlier conditional and 27/32 versions. The proofs have been developed and checked with AI assistance and exact-arithmetic controls. No independent mathematical review or complete Lean verification is claimed.

The proof can be read in the following order.

Part	Role
Sections 2–3	Formal words, exact certificates, and single-floor counts
Sections 4.2–4.3	Four-step estimate and the *OOEOE* split
Section 4.4 and Appendix C	The precise *OOOEE* mixed-mode theorem
Appendix A	Floor exceptions, positive Fourier errors, and exact carries
Appendix B	The wave-bearing estimate used in Appendix C
Sections 5–6	Five-step density and the conditional all-depth implication

2. Exact itineraries and the power envelope

Throughout, $e(t) = \exp(2\pi i t)$, $\{t\} = t - \lfloor t \rfloor$, and $\|t\|$ is the distance from t to the nearest integer. Define

$$\psi(t) = (-1)^{\lfloor t \rfloor} = 1 - 2\mathbf{1}_{[1/2,1)}(\{t/2\}).$$

Thus ψ equals $+1$ at an even floor and -1 at an odd floor, including at integer endpoints. Implied constants are independent of the scale and of modes in the stated ranges; fixed comparison constants and any displayed ε may enter them.

Let $w = w_1 \cdots w_d$ be a word in $\{E, O\}$. The itinerary $\mathrm{word}_d(n)$ records the parities of $n, J(n), \ldots, J^{d-1}(n)$. It contains d letters and specifies the d operations that produce $J^d(n)$. Write $o(w)$ for the number of its odd letters.

Proposition 2.1 (power envelope). If w is realized at n, then

$$\left(J^d(n)\right)^{2^d} \le n^{3^{o(w)}}.$$

Consequently, if $n \ge 2$ and $3^{o(w)} < 2^d$, then $J^d(n) < n$.

Proof. At any positive integer x, the even operation satisfies $J(x)^2 \le x$, and the odd operation satisfies $J(x)^2 \le x^3$. Suppose a prefix of length r with o odd letters ends at x and obeys $x^{2^r} \le n^{3^o}$. An even operation preserves the right-hand exponent after raising $J(x)^2 \le x$ to 2^r; an odd operation replaces it by 3^{o+1}. Induction starts at the empty prefix. If $3^{o(w)} < 2^d$ and $n > 1$, the bound is strictly less than n^{2^d}, proving descent. □

A word satisfying $3^{o(w)} < 2^{|w|}$ is called *contracting*. A realized contracting prefix is a *power-envelope descent certificate*. Let

$$\mathcal{C}_d = \{n \ge 2 : \text{some contracting prefix of length at most } d \text{ is realized at } n\}.$$

Also set

$$\mathcal{C}_\infty = \bigcup_{d\ge 1} \mathcal{C}_d.$$

These sets refer to this particular sufficient certificate. Flooring can cause descent even when the envelope does not certify it. Therefore $\mathcal{C}_d$ need not equal the set of all starts that descend within d steps.

For an odd-rooted target word w, define its *formal chain* for every odd n, regardless of whether n realizes w:

$$x_1^w(n) = n, \qquad \xi_t^w(n) = \left(x_t^w(n)\right)^{p_t}, \qquad x_{t+1}^w(n) = \lfloor \xi_t^w(n) \rfloor \quad (1 \le t < d),$$

where $p_t = 3/2$ for $w_t = O$ and $p_t = 1/2$ for $w_t = E$. Set $s_t^w(n) = \psi(\xi_t^w(n))$, and put $\epsilon_t = 1$ for $w_{t+1} = E$, $\epsilon_t = -1$ for $w_{t+1} = O$.

Lemma 2.2 (parity and branch indicators). For real x, $\lfloor x \rfloor$ is odd exactly when $\{x/2\} \in [1/2, 1)$. For every odd n and odd-rooted word w,

$$\mathbf{1}_{\mathrm{word}_d(n)=w} = 2^{-(d-1)} \prod_{t=1}^{d-1} (1 + \epsilon_t s_t^w(n)). \tag{2.1}$$

Proof. Writing $x = \lfloor x \rfloor + \{x\}$ proves the first assertion, including the half-open endpoints. Each factor divided by two tests the parity of x_{t+1}^w. Until the first failed test, the formal chain agrees with the true orbit, by induction on t. A failed test makes the product zero and prevents realization of w. If all tests pass, the chains agree through the required letters and the product is one. The empty product covers $d = 1$. □

The use of formal chains is essential: the product is defined on every odd input before any correlation or Fourier expansion is made.

3. An unconditional certificate density

Let $M(N) = \#\{n \le N : n \text{ odd}\} = N/2 + O(1)$, and set

$$S_O(N) = \sum_{\substack{n \le N \\ n \text{ odd}}} \psi(n^{3/2}).$$

Theorem 3.1 (single-floor parity). One has

$$S_O(N) = O(N^{5/6}).$$

In particular,

$$\#\{n \le N : \mathrm{word}_2(n) = OO\} = N/4 + O(N^{5/6}),$$
$$\#(\mathcal{C}_2 \cap [1, N]) = 3N/4 + O(N^{5/6}).$$

Proof. Write $n = 2r + 1$ and $g(r) = \frac{1}{2}(2r+1)^{3/2}$. Then $g''(r) = \frac{3}{2}(2r+1)^{-1/2}$. On a dyadic block with $r \asymp Q$, the classical second-derivative test [2] gives

$$\left| \sum_{r \asymp Q} e(hg(r)) \right| \ll h^{1/2} Q^{3/4} + h^{-1/2} Q^{1/4} \quad (h \ge 1).$$

The Erdős–Turán inequality [3, Chapter 2] bounds the unnormalized interval discrepancy by

$$\ll Q/H + \sum_{h=1}^{H} \frac{1}{h} \left| \sum_{r \asymp Q} e(hg(r)) \right| \ll Q/H + H^{1/2} Q^{3/4} + Q^{1/4}.$$

Taking $H = \lfloor Q^{1/6} \rfloor$ for sufficiently large Q gives $O(Q^{5/6})$. Lemma 2.2 identifies parity with membership in $[1/2, 1)$; summing the dyadic bounds proves the first assertion. Since $S_O = M - 2\#OO$, it also proves the count of OO.

The only minimal contracting words of length at most two are E and OE. The first class has count $\lfloor N/2 \rfloor$, and the second has count $M - \#OO = N/4 + O(N^{5/6})$. They are disjoint, and Proposition 2.1 proves the stated certificate count. □

This argument estimates starting values in an interval. It makes no independence assertion about successive orbit parities.

Proposition 3.2 (the OE third letter). One has

$$\#\{n \le N : \mathrm{word}_3(n) = w\} = N/8 + O(N^{5/6} \log(2N)), \qquad w \in \{OEE, OEO\}.$$

Proof. For every real $x \ge 0$,

$$\lfloor \sqrt{\lfloor x \rfloor} \rfloor = \lfloor \sqrt{x} \rfloor :$$

for an integer $r \ge 0$, the inequalities $r^2 \le \lfloor x \rfloor$ and $r^2 \le x$ are equivalent. Consequently, with $m = \lfloor n^{3/2} \rfloor$, one has $\psi(m^{1/2}) = \psi(n^{3/4})$ exactly.

On a dyadic block apply the two-dimensional Erdős–Turán–Koksma inequality to $(\{n^{3/2}/2\}, \{n^{3/4}/2\})$ at odd n, with cutoff $H = \lfloor P^{1/6} \rfloor$. For a nonzero integer mode (i, l) in this range, the second-derivative estimate gives

$$\Big|\sum e\big(\tfrac{i}{2} n^{3/2} + \tfrac{l}{2} n^{3/4}\big)\Big| \ll \begin{cases} |i|^{1/2} P^{3/4} + |i|^{-1/2} P^{1/4}, & i \ne 0, \\ |l|^{1/2} P^{3/8} + |l|^{-1/2} P^{5/8}, & i = 0. \end{cases}$$

For $i \ne 0$, dominance follows from $|l/i| P^{-3/4} \le P^{-7/12}$. Using the product weights of Proposition 4.2, the discrepancy is

$$O\big(P/H + H^{1/2} P^{3/4} \log(2H) + P^{5/8} \log(2H)\big),$$

hence $O(P^{5/6} \log P)$. The two boxes selecting even m and either parity of $\lfloor n^{3/4} \rfloor$ have area $1/4$. Each therefore counts $P/8$ odd starts on the block, with the stated error. Sum dyadically. □

4. Correlation criteria and proved cases

For a fixed odd-rooted word w of length d and nonempty $A \subseteq \{1, \ldots, d-1\}$, define

$$R_{w,A}(N) = \sum_{\substack{n \le N \\ n \text{ odd}}} \prod_{t \in A} s_t^w(n).$$

Hypothesis $\mathrm{H}(w; \delta)$**.** For some fixed $0 < \delta < 1$, every nonempty A satisfies $R_{w,A}(N) = O_{w,A}(N^{1-\delta})$ as $N \to \infty$.

Hypothesis $\mathrm{H}_0(w)$**.** The same correlations satisfy $R_{w,A}(N) = o(N)$, without a prescribed rate.

These are assertions about the formal chains, not assumptions that the true orbit is a sequence of independent coin tosses. Hypothesis $\mathrm{H}(w; \delta)$ implies $\mathrm{H}_0(w)$.

Proposition 4.1 (conditional word count). Under $\mathrm{H}(w;\delta)$,

$$\#\{n \le N : \mathrm{word}_d(n) = w\} = 2^{-d}N + O_w(N^{1-\delta}). \tag{4.1}$$

Under $\mathrm{H}_0(w)$, the error is $o(N)$.

Proof. Expand (2.1) and sum. The empty subset contributes $2^{-(d-1)}M(N) = 2^{-d}N + O(1)$. The other $2^{d-1} - 1$ terms are fixed signed multiples of $R_{w,A}$, and there are finitely many for fixed w. □

For example, $\mathrm{H}(OOEE;\delta)$ concerns the seven nonempty products of

$$\psi(n^{3/2}), \qquad \psi(m^{3/2}), \qquad \psi(v^{1/2}), \quad m = \lfloor n^{3/2} \rfloor, \quad v = \lfloor m^{3/2} \rfloor.$$

Proposition 4.1 then gives the count $N/16 + O(N^{1-\delta})$ needed for four-step certificates. Theorem 3.1 establishes the first estimate. Corollary 4.6 below establishes all seven for every $0 < \delta < 1/24$.

4.1 A sufficient Fourier criterion

For $\mathbf{k} = (k_1, \ldots, k_{d-1}) \in \mathbb{Z}^{d-1}$, let

$$S_{w,\mathbf{k}}(N) = \sum_{\substack{n \le N \\ n \text{ odd}}} e\left(\frac{1}{2}\sum_{t=1}^{d-1} k_t \xi_t^w(n)\right).$$

Proposition 4.2 (conditional Fourier-to-parity transfer). Fix $d \ge 2$, w, and $0 < \eta, \sigma < 1$. Suppose, uniformly over all nonzero integer vectors with $\|\mathbf{k}\|_\infty \le \lfloor N^\eta \rfloor$, that

$$|S_{w,\mathbf{k}}(N)| \ll_w N^{1-\sigma}. \tag{4.2}$$

Then $\mathrm{H}(w;\delta)$ holds for every $0 < \delta < \min(\eta, \sigma)$. The constants may depend on these fixed parameters.

Proof. Apply the multidimensional Erdős–Turán–Koksma inequality [3, Chapter 2, p. 116] to the points $\mathbf{y}_n = (\{\xi_1^w(n)/2\}, \ldots, \{\xi_{d-1}^w(n)/2\})$. For any half-open axis-parallel box $B \subseteq [0,1)^{d-1}$, its unnormalized discrepancy is

$$|\#\{n \le N \text{ odd} : \mathbf{y}_n \in B\} - M(N)|B|| \ll_d \frac{M(N)}{H} + \sum_{0 < \|\mathbf{k}\|_\infty \le H} \frac{|S_{w,\mathbf{k}}(N)|}{r(\mathbf{k})},$$

$$r(\mathbf{k}) = \prod_{t=1}^{d-1} \max(1, |k_t|).$$

The weight sum is $O_d((1+\log H)^{d-1})$. With $H = \lfloor N^\eta \rfloor$, (4.2) gives box discrepancy $O_w(N^{1-\eta} + N^{1-\sigma}(1+\log N)^{d-1})$. Each nonempty sign product is constant, with value 1 or -1, on the 2^{d-1} boxes obtained by splitting each coordinate at $1/2$. Its integral over the unit cube is zero. Summing their discrepancies therefore bounds $R_{w,A}$; absorbing the fixed logarithmic power proves the assertion. □

The criterion needs every indicated mixed mode, including vectors with zero coordinates and mixed signs. A bound for one undecorated kernel is not a substitute for (4.2). Qualitatively, cancellation $S_{w,\mathbf{k}}(N) = o(N)$ for each fixed nonzero $\mathbf{k}$ also implies $\mathrm{H}_0(w)$: keep H fixed in the displayed inequality, let $N \to \infty$, and then let $H \to \infty$.

4.2 A nested estimate sufficient for four-step descent

The general decorated kernel is not needed for four-step certificates. We prove the smaller mixed family directly. Constants in this subsection may depend on a fixed $C \ge 1$, but are uniform in all integer modes in their stated ranges.

Lemma 4.3 (a truncated carry expansion). Put

$$b(t)=\{t\}-\tfrac{1}{2},\qquad b_R(t)=-\sum_{1\le|r|\le R}\frac{e(rt)}{2\pi ir},\qquad E_R(t)=\min\left(1,\frac{1}{R\|t\|}\right),$$

where $\|t\|$ denotes distance to the nearest integer, $E_R(t)=1$ at integers, and $R\ge 2$ is integral. Then $b(t)=b_R(t)+O(E_R(t))$, including at integers. For $X(x)=x^{3/2}$, any interval $I\subseteq[P,3P]$ satisfies

$$\sum_{\substack{n\in I\\ n\text{ odd}}}E_R(X(n))\ll\frac{P\log(2R)}{R}+P^{5/6}.\tag{4.3}$$

Proof. Away from integers, the Fourier series of b is the displayed series with infinite range. Summation by parts bounds its tail by $O((R\|t\|)^{-1})$. If $\|t\|\le R^{-1}$, pairing the modes r and $-r$ and using $|\sin(2\pi rt)|\le 2\pi r\|t\|$ bounds the finite sum by an absolute constant. At an integer the finite sum is zero and $b(t)=-1/2$, so the asserted bound remains valid.

The proof of Theorem 3.1, with phase $(2r+1)^{3/2}$, gives interval discrepancy $O(P^{5/6})$ for the fractional parts $\{X(n)\}$ with odd $n\in I$. In particular the number within distance $z\le 1/2$ of an integer is $O(Pz+P^{5/6})$. Split these distances at $R^{-1},2R^{-1},4R^{-1},\ldots$. Each resulting weighted count contributes $O(P/R+2^{-j}P^{5/6})$, proving (4.3). This is a bound in terms of the ambient scale P; it does not rescale a global exceptional set by the length of I. □

Lemma 4.4 (small-shift nested sum). Write $m(n)=\lfloor n^{3/2}\rfloor$, $Y(n)=m(n)^{3/2}$, and $\Delta_hf(n)=f(n+2h)-f(n)$. If

$$1\le h\le P^{1/12},\qquad 1\le|j|\le CP^{1/24},\qquad |i|,|k|\le CP^{1/24},$$

where i,j,k,h are integers, then

$$\left|\sum_{\substack{P<n\le 2P-2h\\ n\text{ odd}}}e\left(\tfrac{i}{2}\Delta_hX(n)+\tfrac{j}{2}\Delta_hY(n)+\tfrac{k}{2}\Delta_h(n^{9/8})\right)\right|\ll_C P^{7/8}(1+h^{1/2}).\tag{4.4}$$

Proof. Conjugate the whole sum if necessary and put $u=j/2>0$. Thus $u\ge 1/2$; integrality of u is not required. Set $\theta=\{X(n)\}$, $\delta=\Delta_hX(n)$, and $g=m(n+2h)-m(n)$. Lemma 7.1 gives the exact identity

$$\Delta_hY=A_h+\tfrac{3}{2}g(n+2h)^{3/4}-\tfrac{3}{2}\theta\Delta_h(n^{3/4})+\Delta_hE,\tag{4.5}$$

where

$$A_h(x)=\tfrac{3}{2}x^{3/2}\Delta_h(x^{3/4})-\tfrac{1}{2}\Delta_h(x^{9/4}),\qquad A_h''(x)=O(h^2P^{-7/4}).$$

For the derivative assertion, write $A_h(x)=x^{9/4}a(2h/x)$, where $a(z)=\frac{3}{2}((1+z)^{3/4}-1)-\frac{1}{2}((1+z)^{9/4}-1)$. The identities $a(0)=a'(0)=0$ and bounded derivatives of $a(z)/z^2$ near zero give the estimate. No floor is differentiated.

Deleting the last two terms of (4.5) from the phase costs

$$O(uhP^{3/4}+uP^{1/4}),\tag{4.6}$$

because $\Delta_h(x^{3/4})=O(hP^{-1/4})$, $E(x)=O(P^{-3/4})$, and $|e(t)-1|\le 2\pi|t|$.

Partition the interval into level sets of $G=\lfloor\delta\rfloor$. Since $\delta'(x)\asymp hP^{-1/2}$, there are $O(1+hP^{1/2})$ cells; their total length is at most P. On a cell put $z=\delta-G$. It is monotone in $[0,1)$, and exactly

$$g=G+\kappa,\qquad \kappa=z+b(X(n))-b(X(n+2h))\in\{0,1\}.\tag{4.7}$$

Define the two smooth phases on this cell by

$$F_{G,\epsilon}(x)=uA_h(x)+\tfrac{3u}{2}(G+\epsilon)(x+2h)^{3/4}+\tfrac{i}{2}\delta(x)+\tfrac{k}{2}\Delta_h(x^{9/8}),\quad \epsilon\in\{0,1\}.$$

The exponential left after (4.6) is exactly

$$\begin{aligned}&(1-z)e(F_{G,0})+ze(F_{G,1})\\&\quad+\big(b(X(n))-b(X(n+2h))\big)\big(e(F_{G,1})-e(F_{G,0})\big).\end{aligned}\tag{4.8}$$

Apply Lemma 4.3 with $R=\lfloor P^{1/4}\rfloor$ to the two sawtooths. Their total error after the partition is $O(P^{5/6}+P^{3/4}\log P)$: the cells partition the same input set, and the multiplying difference of exponentials has modulus at most two.

On every cell $G+\epsilon\asymp hP^{1/2}$, and

$$F''_{G,\epsilon}(x)=-\tfrac{9}{32}u(G+\epsilon)(x+2h)^{-5/4}+O(uh^2P^{-7/4}+|i|hP^{-3/2}+|k|hP^{-15/8}).\tag{4.9}$$

The error is $o(uhP^{-3/4})$, uniformly in the stated ranges. The second-derivative estimate [2, Theorem 2.2] has scale $\lambda\asymp uhP^{-3/4}$ and a bounded curvature ratio. After $n=2r+1$ it has the same form for odd integers, with implicit constants. Summing over all cells, and using partial summation for the monotone weights z and $1-z$, gives

$$\ll (uh)^{1/2}P^{5/8}+(h/u)^{1/2}P^{7/8}.\tag{4.10}$$

Each nonzero Fourier mode adds either $rX(x)$ or $rX(x+2h)$ to one of the phases $F_{G,\epsilon}$. Its curvature has size $|r|P^{-1/2}$, which dominates (4.9), since

$$\frac{uhP^{-3/4}}{|r|P^{-1/2}}\ll_C P^{-1/8}\qquad(1\le|r|\le R).$$

There is no curvature-collision regime in this proof. Summing the second-derivative estimate over all gap cells costs, per mode,

$$\ll |r|^{1/2}P^{3/4}+h|r|^{-1/2}P^{3/4}.$$

The Fourier weights are $O(1/|r|)$, so their total is $O(R^{1/2}P^{3/4}+hP^{3/4})$. Collecting the costs gives

$$\begin{aligned}&\ll_C (uh)^{1/2}P^{5/8}+(h/u)^{1/2}P^{7/8}+P^{7/8}+hP^{3/4}\\&\qquad+uhP^{3/4}+uP^{1/4}+P^{5/6}+P^{3/4}\log P.\end{aligned}$$

Here $uh\ll_C P^{1/8}$ and $u\ge 1/2$. Every term is $O_C(P^{7/8}(1+h^{1/2}))$. Endpoint cells are included; no estimate is extended across their boundaries. □

Theorem 4.5 (restricted mixed exponential sums). For every fixed $C\ge1$, uniformly over nonzero integer triples (i,j,k) with $\max(|i|,|j|,|k|)\le CP^{1/24}$, one has

$$\left|\sum_{\substack{P<n\le 2P\\ n\text{ odd}}} e\left(\tfrac{i}{2}n^{3/2}+\tfrac{j}{2}m(n)^{3/2}+\tfrac{k}{2}n^{9/8}\right)\right|\ll_C P^{23/24}.\tag{4.11}$$

Proof. If $j\ne0$, apply the van der Corput differencing inequality [2] to the sequence indexed by odd integers, with $H=\lfloor P^{1/12}\rfloor$. Writing T_h for (4.4), it gives

$$|S|^2\ll\frac{P^2}{H}+\frac{P}{H}\sum_{1\le h<H}|T_h|\ll_C\frac{P^2}{H}+P^{15/8}H^{1/2}\ll_C P^{23/12}.$$

If $j=0$, the phase is smooth. For $i\ne0$, its curvature is $\asymp|i|P^{-1/2}$, because $|k/i|P^{-3/8}\ll_C P^{-1/3}$. The bound is $O_C(P^{37/48}+P^{1/4})$. For $i=0$, one has $k\ne0$, curvature $\asymp|k|P^{-7/8}$, and bound $O_C(P^{7/12}+P^{7/16})$. Both are smaller than (4.11). □

Corollary 4.6 (three formal signs). Put $v = \lfloor m^{3/2} \rfloor$. Every sign class of

$$\big(\psi(n^{3/2}), \psi(m^{3/2}), \psi(v^{1/2})\big)$$

on odd $n \le N$ has count

$$\frac{N}{16} + O\big(N^{23/24}(\log(2N))^3\big). \tag{4.12}$$

In particular, $\mathrm{H}(OOEE;\delta)$ holds for every $0 < \delta < 1/24$.

Proof. On a dyadic block,

$$|v^{1/2} - n^{9/8}| \le |v^{1/2} - m^{3/4}| + |m^{3/4} - X^{3/4}| \ll P^{-9/8} + P^{-3/8}.$$

Replacing the third phase coordinate in (4.11) by $v^{1/2}$ costs $O_C(|k|P^{5/8}) = O_C(P^{2/3})$. Apply the three-dimensional Erdős–Turán–Koksma inequality, as in Proposition 4.2, to the points

$$\big(\{X/2\}, \{Y/2\}, \{v^{1/2}/2\}\big),$$

with cutoff $\lfloor P^{1/24} \rfloor$ on this block. Its box discrepancy is $O(P^{23/24}(\log(2P))^3)$. The eight half-cube boxes encode the three signs exactly. Summing over dyadic blocks proves (4.12) and bounds all seven nonempty sign products by the same error. The cutoff is chosen separately on each block; no uniformity from a large block is assumed on smaller ones. □

This establishes the hypothesis used for four-step certificates. It does not estimate the expanding third-level coordinate $v^{3/2}$ or every formal chain required at length five. The next subsection handles the $OOEOE$ chain.

4.3 The fifth-letter split of OOEO

Write

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad U = v^{1/2}, \quad w = \lfloor U \rfloor, \quad W = w^{3/2}.$$

The four formal coordinates for $OOEOE$ are X, Y, U, W. The new floor is handled by centering its Fourier expansion at the integer part of a smooth coefficient. This introduces many intervals; the next two lemmas keep their boundary cost explicit.

Lemma 4.7 (a centered Fourier expansion). For $0 \le \beta \le 1$, put

$$a_r(\beta) = \int_0^1 e(-(\beta + r)t)\, dt.$$

For integral $T \ge 2$, uniformly in real t and β,

$$e(-\beta\{t\}) = \sum_{|r| \le T} a_r(\beta) e(rt) + O(E_T(t)), \qquad |a_r(\beta)| + |a_r'(\beta)| \ll (1 + |r|)^{-1}. \tag{4.13}$$

Here E_T is as in Lemma 4.3. If $B = N + \beta$, $N \in \mathbb{Z}$, then $e(-B\{t\}) = e(-Nt)e(-\beta\{t\})$ exactly. On an interval where $N = \lfloor B(x) \rfloor$ is fixed and $\beta(x)$ is monotone, the sum over $|r| \le T$ of the supremum norms and total variations of $a_r(\beta(x))$ is $O(\log(2T))$.

Proof. The integral defines the coefficients also at the removable singularities of

$$a_r(\beta) = \frac{1 - e(-\beta)}{2\pi i(r + \beta)}.$$

Integration by parts gives the asserted bounds for $|r| \ge 2$, including for the derivative with respect to β; the remaining coefficients and their derivatives are bounded by their integrals. For $|r| \ge 2$, the coefficient is $(1 - e(-\beta))/(2\pi i r) + O(r^{-2})$, uniformly in β. The tail of the first series is bounded by the argument of Lemma 4.3, and the absolutely convergent remainder has tail $O(T^{-1})$. Near an integer the paired $r, -r$ terms are bounded as in that lemma. At the integer itself both sides are bounded and $E_T = 1$. This proves (4.13) at every point. Finally the variation of β on one interval is at most one, and summing the coefficient bounds gives the logarithm. □

Bounded-residual extension. The same expansion holds for every real β with $|\beta| \le B_0$, where B_0 is fixed, with constants depending only on B_0. Moreover, for a real function $\beta(x)$ on an interval I, the condition $\|\beta\|_{\infty,I} + \mathrm{TV}_I(\beta) \le B_0$ implies

$$\sum_{|r|\le T} (\|a_r(\beta)\|_{\infty,I} + \mathrm{TV}_I(a_r \circ \beta)) \ll_{B_0} \log(2T).$$

Monotonicity is not required for this extension.

Proof. For $|r| > 2B_0 + 2$, the integral formula gives $a_r(\beta) = (1 - e(-\beta))/(2\pi i r) + O_{B_0}(r^{-2})$, and $|a_r| + |a_r'| \ll_{B_0} (1 + |r|)^{-1}$. The remaining finitely many coefficients and derivatives are bounded directly by their integrals, including at $r + \beta = 0$. The same paired-tail argument proves the pointwise remainder; bounded cutoffs can be absorbed in the constant since $E_T(t) \ge \min(1, 2/T)$. Finally $\mathrm{TV}_I(a_r \circ \beta) \le \sup_{|v|\le B_0} |a_r'(v)|\,\mathrm{TV}_I(\beta)$. Sum the harmonic bounds. Complex conjugation gives the expansion with the opposite sign in the exponent. □

Lemma 4.8 (mixed sums over the frequency intervals). Set

$$J_0 = P^{1/48}, \qquad H = \lfloor P^{1/12} \rfloor.$$

Let $1 \le |k| \le J_0$, $1/2 \le u \le J_0/2$, and $|i| \le J_0$. Partition a subinterval of $[P, 2P]$ into D intervals I, with

$$D \ll |k| P^{9/16}, \qquad |I| \ll L := P^{7/16}/|k|.$$

For a fixed constant C, define

$$M_I = \sup_{\substack{|R|\le C|k|P^{9/16} \\ I' \subseteq I}} \left| \sum_{\substack{n \in I' \\ n \text{ odd}}} e\left(\tfrac{i}{2} X(n) + uY(n) + \tfrac{k}{2} n^{27/16} + R n^{9/8}\right) \right|.$$

The second supremum ranges over subintervals. Then

$$\sum_I M_I \ll_C P^{23/24}. \tag{4.14}$$

The constants in the partition assumptions are fixed. In particular, this is a bound after summing all the frequency intervals, not a rescaling of a full-block discrepancy estimate.

Proof. Fix a shift $1 \le h < H$. The exact carry argument of Lemma 4.4 applies on every I. The additional smooth part has differenced second derivative

$$O\big(|i| h P^{-3/2} + |k| h P^{-21/16} + |R| h P^{-15/8}\big) = O_C\big(J_0 h P^{-3/2} + |k| h P^{-21/16}\big).$$

This is $o(uhP^{-3/4})$. Thus the zero-mode curvature and the dominance of every nonzero carry mode are unchanged.

Intersecting the frequency intervals with the gap cells produces $O(D + hP^{1/2})$ cells in total: on an interval of length ℓ, the gap function has variation $O(hP^{-1/2}\ell)$. Retaining these boundaries, the zero carry modes cost

$$\ll (uh)^{1/2} P^{5/8} + (h/u)^{1/2} P^{7/8} + D (uh)^{-1/2} P^{3/8}.$$

With carry cutoff $R_0 = \lfloor P^{1/4} \rfloor$, the nonzero modes cost

$$\ll R_0^{1/2} P^{3/4} + h P^{3/4} + D P^{1/4}.$$

These follow by applying the second-derivative estimate on every intersection, exactly as in (4.10). The weights $z, 1-z$ in the carry identity have uniformly bounded variation on each intersection.

The discarded floor terms cost $O(uhP^{3/4}+uP^{1/4})$. The truncated carry error costs $O(P^{5/6}+P^{3/4}\log P)$ over the whole partition. For each local sum, its two endpoints $n, n+2h$ lie in the same I, so this error is bounded by a fixed multiple of $\sum_{n\in I} E_{R_0}(X(n))$. The disjoint intervals therefore charge (4.3) only once. The same bounds hold after taking the suprema over R and subintervals, because their local bounds use only $|I|$, its gap-cell count, and this nonnegative error sum.

If B_h denotes the sum over I of those supremum bounds for the differenced sums, the collected costs imply

$$B_h \ll_C P^{7/8}(1+h^{1/2}) + |k|P^{15/16}h^{-1/2}. \tag{4.15}$$

Indeed $uh \le P^{5/48}/2$, $u \ge 1/2$, and $DP^{1/4} \ll |k|P^{13/16} \le P^{5/6}$.

Apply van der Corput's inequality to each local sum, padding by zero inside an interval of length $O(L)$ when taking a subinterval. Since $H \ll L$, its squared bound is

$$O\left(\frac{L}{H}\left(\#I + \sum_{h<H} |T_{I,h}|\right)\right),$$

where $\#I$ counts odd integers and endpoint constants are harmless. Cauchy–Schwarz over the D intervals, together with $DL \ll P$, gives

$$\begin{aligned}\left(\sum_I M_I\right)^2 &\ll_C \frac{P^2}{H} + \frac{P}{H}\sum_{h<H} B_h \\ &\ll_C \frac{P^2}{H} + P^{15/8}H^{1/2} + |k|P^{31/16}H^{-1/2} \ll_C P^{23/12}.\end{aligned}$$

The three largest exponents are respectively $2-1/12$, $15/8+1/24$, and $1/48+31/16-1/24$, all equal to $23/12$. This proves (4.14), including partial endpoint intervals. □

Theorem 4.9 (four formal coordinates). Uniformly over nonzero integer quadruples with $\max(|i|,|j|,|\ell|,|k|) \le P^{1/48}$,

$$\left|\sum_{\substack{P<n\le 2P \\ n \text{ odd}}} e\left(\tfrac{i}{2}X + \tfrac{j}{2}Y + \tfrac{\ell}{2}U + \tfrac{k}{2}W\right)\right| \ll P^{23/24}\log(2P). \tag{4.16}$$

Proof. We first bound the error in (4.13) along $U(n)$. The comparison $U = n^{9/8} + O(P^{-3/8})$ from Corollary 4.6 gives, for positive integer q,

$$\left|\sum e(qU)\right| \ll q^{1/2}P^{9/16} + q^{-1/2}P^{7/16} + qP^{5/8}.$$

Erdős–Turán with cutoff $\lfloor P^{1/8}\rfloor$ gives interval discrepancy $O(P^{7/8})$ for $\{U(n)\}$. The distance-strip argument of Lemma 4.3 therefore gives

$$\sum E_T(U(n)) \ll P\log(2T)/T + P^{7/8}. \tag{4.17}$$

All sums here are over the same odd-input block.

When $k=0$, Theorem 4.5 and the comparison for U prove (4.16). Suppose $k \ne 0$, and put $\theta = \{X\}$, $\xi = \{U\}$. Taylor's theorem, with its second derivative bounded on the unit floor intervals, gives

$$\begin{aligned} U &= n^{9/8} - \tfrac{3}{4}\theta n^{-3/8} + O(P^{-9/8}), \\ W &= n^{27/16} - \tfrac{9}{8}\theta n^{3/16} - \tfrac{3}{2}n^{9/16}\xi + O(P^{-9/16}). \end{aligned} \tag{4.18}$$

For clarity, $U = m^{3/4} + O(P^{-9/8})$. Also $W = v^{3/4} - (3/2)v^{1/4}\xi + O(P^{-9/16})$, $v^{3/4} = m^{9/8} + O(P^{-9/16})$, and $v^{1/4} = n^{9/16}+O(P^{-15/16})$; expanding $m = X-\theta$ proves both displayed formulas.

Set
$$B(x) = \tfrac{3k}{4}x^{9/16}, \qquad C(x) = \tfrac{9k}{16}x^{3/16}, \qquad T = \lfloor P^{1/8}\rfloor.$$

Partition by $N = \lfloor B(x)\rfloor$, and put $\beta = B - N$. Since $|B'| \asymp |k|P^{-7/16}$, this partition has $D \ll |k|P^{9/16}$ intervals, each of length $O(P^{7/16}/|k|)$. The two endpoint intervals can be shorter. The total variation of β on any one interval is at most one.

Use (4.18), then expand $e(-B\{U\})$ by Lemma 4.7. Formula (4.17) bounds its total truncation error by $O(P^{7/8}\log P)$; the initial Taylor error is $O(|k|P^{7/16})$. On a fixed interval and for $|r| \le T$, set
$$R = r - N + \ell/2.$$

This is a fixed frequency on that interval. The corresponding phase is
$$\tfrac{i}{2}X + \tfrac{j}{2}Y + \tfrac{k}{2}n^{27/16} + RU - C\theta.$$

The first formula in (4.18) replaces it by
$$\tfrac{i}{2}X + \tfrac{j}{2}Y + \tfrac{k}{2}n^{27/16} + Rn^{9/8} - \big(C + \tfrac{3}{4}Rn^{-3/8}\big)\theta + O(|R|P^{-9/8}).$$

The essential cancellation is the exact identity
$$C + \tfrac{3}{4}Rn^{-3/8} = \tfrac{3}{4}n^{-3/8}(B + R) = \tfrac{3}{4}n^{-3/8}(r + \beta + \ell/2). \tag{4.19}$$

Its absolute value is $O((T + P^{1/48})P^{-3/8})$, so the remaining θ -term can be deleted. Since $|R| \ll |k|P^{9/16}$, all these replacements, summed with the coefficient masses, cost
$$O\big((|k|P^{7/16} + P^{5/8}(T + P^{1/48}))\log P\big) \ll P^{3/4}\log P.$$

Thus it remains to estimate the smooth-frequency phases in Lemma 4.8. If $j \neq 0$, conjugation if needed makes $u = |j|/2 \ge 1/2$. Partial summation for the coefficients $a_r(\beta)$, whose summed supremum norms and variations are $O(\log P)$ on each interval, and (4.14) give $O(P^{23/24}\log P)$.

If $j = 0$, differentiate with R fixed. The remaining phase $\Phi = iX/2 + kn^{27/16}/2 + Rn^{9/8}$ satisfies
$$\begin{aligned}\Phi''(x) &= \tfrac{3i}{8}x^{-1/2} + \tfrac{297k}{512}x^{-5/16} + \tfrac{9R}{64}x^{-7/8}\\ &= \tfrac{243k}{512}x^{-5/16} + O\big(|i|P^{-1/2} + (T + |\ell| + 1)P^{-7/8}\big).\end{aligned}$$

The last equality substitutes $R = -B(x) + (r + \beta(x) + \ell/2)$ only after differentiation. The error is $o(|k|P^{-5/16})$. Summing the second-derivative bounds over every frequency interval costs
$$O\big(|k|^{1/2}P^{27/32} + D|k|^{-1/2}P^{5/32}\big) \ll |k|^{1/2}(P^{27/32} + P^{23/32}).$$

The logarithmic coefficient mass still leaves this below (4.16). All zero coordinates and both signs of every mode have now been covered. □

Corollary 4.10 (the OOEO split). Each of the sixteen formal sign classes of $(\psi(X), \psi(Y), \psi(U), \psi(W))$ over odd $n \le N$ has count

$$N/32 + O(N^{47/48}). \tag{4.20}$$

In particular, for each $a \in \{OOEOE, OOEOO\}$,

$$\#\{n \le N : \mathrm{word}_5(n) = a\} = N/32 + O(N^{47/48}).$$

The hypothesis $\mathrm{H}(OOEOE; 1/48)$ holds.

Proof. Apply four-dimensional Erdős–Turán–Koksma on each dyadic block with cutoff $\lfloor P^{1/48} \rfloor$. Theorem 4.9 bounds its box discrepancy by

$$O\big(P^{47/48} + P^{23/24}(\log(2P))^5\big) = O(P^{47/48}).$$

The sixteen half-cubes have volume 1/16; the number of odd inputs is $P/2 + O(1)$. Summing dyadically proves (4.20). All fifteen nonempty sign products are bounded by summing these discrepancies. The signs $(-1, +1, -1, \pm 1)$ select the two actual words by Lemma 2.2. This does not count the formal chain through the expanding third-level coordinate $v^{3/2}$. □

4.4 The OOOEE mixed-mode estimate

For this subsection the formal coordinates are

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad Z = v^{3/2}, \quad w = \lfloor Z \rfloor, \quad U = \sqrt{w}.$$

They differ from the chain in Section 4.3.

Theorem 4.11 (OOOEE mixed modes). For every fixed $C \ge 1$ and every $\varepsilon > 0$, uniformly over nonzero integer quadruples with $\max(|i|, |j|, |k|, |l|) \le CP^{1/24}$,

$$\left| \sum_{\substack{P < n \le 2P \\ n \text{ odd}}} e\left(\tfrac{1}{2}(iX + jY + kZ + lU)\right) \right| \ll_{C,\varepsilon} P^{127/128+\varepsilon}. \tag{4.21}$$

Proof. Appendix C provides the full argument using Lemmas A.1–A.4 and Theorem B.1. The cases with $k = 0$ are treated directly. For $k \ne 0$, two differencings use half-shift cutoffs $P^{1/48}$ and $P^{1/24}$. The complete second correlation has exponent 31/32, including all positive errors. The two outer inequalities give exponents 63/64 and 127/128; their diagonal terms are smaller. □

Corollary 4.12 (the OOOE split). Every formal parity-sign class of (X, Y, Z, U) on odd $n \le N$ has count

$$N/32 + O_\varepsilon(N^{127/128+\varepsilon}). \tag{4.22}$$

In particular $OOOEE$ and $OOOEO$ each have that count, and $\mathrm{H}(OOOEE; \delta)$ holds for every $0 < \delta < 1/128$.

Proof. On each dyadic block apply Erdős–Turán–Koksma at cutoff $\lfloor P^{1/24} \rfloor$ to the four coordinates divided by two, as in Proposition 4.2. The sixteen half-cubes all have volume 1/16, and there are $P/2 + O(1)$ odd inputs. Absorb the fixed logarithmic weights in (4.21) into epsilon, then sum dyadically with each block's own cutoff. The signs $(-1, -1, +1, \pm 1)$ select the stated actual words. Summing the sixteen discrepancies also controls every nonempty sign product. The end of Appendix C gives the endpoint conventions and full dyadic argument. □

5. Finite-depth descent densities

Lemma 5.1 (minimal certificates through length five). The contracting words of length at most five with no proper contracting prefix are exactly

$$E, \qquad OE, \qquad OOEE, \qquad OOOEE, \qquad OOEOE. \tag{5.1}$$

Proof. An E -rooted word already contracts at its first letter, and an OE -rooted word contracts at its second. Every remaining word begins with OO. At length three, even two odd letters give $3^2 > 2^3$, so no new certificate appears. At length four, contraction requires at most two odd letters, giving only $OOEE$. After excluding that prefix, the possible fourth-level prefixes are $OOEO$, $OOOE$, and $OOOO$. At length five, contraction requires at most three odd letters because $3^3 < 2^5 < 3^4$. The first two prefixes must therefore end with E, and the third cannot contract. This proves (5.1). □

Theorem 5.2 (unconditional four-step density). One has

$$\#(\mathcal{C}_4 \cap [1, N]) = \frac{13N}{16} + O\big(N^{23/24}(\log(2N))^3\big).$$

Proof. Lemma 5.1 gives the disjoint prefix classes E, OE, and $OOEE$. Theorem 3.1 supplies the first two counts. The $OOEE$ class is the sign class $(-1, +1, +1)$ in Corollary 4.6, so has count $N/16 + O(N^{23/24}(\log(2N))^3)$. Their densities add to $1/2 + 1/4 + 1/16 = 13/16$. □

Theorem 5.3 (a five-step subfamily of density $27/32$). Let

$$\mathcal{D}_5 = \mathcal{C}_4 \ \cup\ \{n \geq 2 : \mathrm{word}_5(n) = OOEOE\}.$$

Then $\mathcal{D}_5 \subseteq \mathcal{C}_5$ and

$$\#(\mathcal{D}_5 \cap [1, N]) = 27N/32 + O(N^{47/48}).$$

In particular the lower natural density of $\mathcal{C}_5$ is at least $27/32$.

Proof. The added prefix is disjoint from $E, OE, OOEE$, and $3^3 < 2^5$ certifies its contraction. Add Corollary 4.10 to Theorem 5.2, absorbing the smaller four-step error. The main terms sum to $13/16 + 1/32 = 27/32$. This does not assert that $\mathcal{D}_5$ exhausts $\mathcal{C}_5$. □

Theorem 5.4 (full five-step certificate density). For every $\varepsilon > 0$,

$$\#(\mathcal{C}_5 \cap [1, N]) = \frac{7N}{8} + O_\varepsilon(N^{127/128+\varepsilon}).$$

In particular $\mathcal{C}_5$ has natural density $7/8$. This counts the power-envelope certificate class, not every start that may happen to descend within five operations.

Proof. By Lemma 5.1, $\mathcal{C}_5$ is the disjoint union of $\mathcal{D}_5$ and the $OOOEE$ class. Add Theorem 5.3 and Corollary 4.12. Their main terms are $27N/32 + N/32 = 7N/8$; the former error $O(N^{47/48})$ is smaller. □

The exact fractions and their analytic requirements can be read together:

Certificate family	Minimal prefixes added	Density	Status
$\mathcal{C}_1$	E	1/2	Unconditional
$\mathcal{C}_2$	OE	3/4	Theorem 3.1
$\mathcal{C}_4$	$OOEE$	13/16	Theorem 5.2
$\mathcal{D}_5 \subseteq \mathcal{C}_5$	$OOEOE$	27/32	Theorem 5.3
$\mathcal{C}_5$	$OOOEE$	7/8	Theorem 5.4

The formal $OOOEE$ chain in Section 4.4 uses the expanding third-level coordinate $v^{3/2}$; the $OOEOE$ chain in Section 4.3 uses $v^{1/2}$ at that level. The proofs therefore concern distinct families. Neither proof asserts fair-share counts for every actual itinerary at every depth.

6. The fixed-depth reduction to density-one certificates

Hypothesis FD. For every fixed $d \geq 1$ and every odd-rooted word w of length d,

$$\#\{n \leq N : \mathrm{word}_d(n) = w\} = 2^{-d}N + o_w(N).$$

No common error rate in d is assumed. The family $\mathrm{H}_0(w)$ over all such words implies FD.

Theorem 6.1 (conditional density-one certificate theorem). Under FD, $\mathcal{C}_\infty$ has natural density one.

Proof. Put $p = \log 2/\log 3 > 1/2$. A word with no contracting prefix through length d must in particular have at least pd odd letters at length d. All such words begin with O. For a uniformly chosen odd-rooted word of length d, the number of odd letters is distributed as $1 + B_{d-1}$, where B_{d-1} is a binomial random variable with parameters $(d-1, 1/2)$. This is an auxiliary count on words, not a stochastic model asserted for the Juggler orbit.

Choose $q = (p + 1/2)/2$, so $1/2 < q < p$. For all sufficiently large fixed d, $pd - 1 \geq q(d-1)$. For $t > 0$, the exponential Markov inequality and the binomial generating function give

$$\Pr(B_{d-1} \geq q(d-1)) \leq \left(\frac{1+e^t}{2e^{qt}}\right)^{d-1}.$$

The logarithm of the expression in parentheses is zero at $t = 0$ and has derivative $1/2 - q < 0$ there. Hence some fixed $t > 0$ makes it a number $\vartheta < 1$.

For fixed d, FD and a finite sum over surviving words show that the natural density of $\mathbb{N} \setminus \mathcal{C}_d$ is their number divided by 2^d. It is at most $\frac{1}{2}\vartheta^{d-1}$ for sufficiently large d. Since $\mathcal{C}_d \subseteq \mathcal{C}_\infty$,

$$\overline{\mathrm{dens}}(\mathbb{N} \setminus \mathcal{C}_\infty) \leq \tfrac{1}{2}\vartheta^{d-1}.$$

Letting $d \to \infty$ proves the assertion. In this argument $N \to \infty$ is taken first, with d fixed. □

The conclusion concerns descent below the initial value. It does not prove that a descended value reaches 1, or that subsequent orbit samples avoid an exceptional set. Even a density-zero exceptional set can contain a nontrivial cycle. No interchange of the two limits, and no assertion of finite stopping time for every start, follows from Theorem 6.1.

7. Exact floor defects and separate analytic questions

For $n \geq 3$, write

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad \theta = X - m, \qquad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad \theta_2 = Y - v.$$

Lemma 7.1 (one-signed linearization). One has

$$m^{3/2} = \tfrac{3}{2}mn^{3/4} - \tfrac{1}{2}n^{9/4} + E(n), \qquad 0 \leq E(n) \leq \tfrac{3}{8}(n^{3/2} - 1)^{-1/2} \leq \tfrac{1}{2}n^{-3/4}.$$

Proof. Let $a = \sqrt{m}$, $b = \sqrt{X}$. Direct factorization gives

$$E = a^3 - \tfrac{3}{2}a^2 b + \tfrac{1}{2}b^3 = \tfrac{1}{2}(a-b)^2(2a+b) \geq 0.$$

Moreover $\theta = b^2 - a^2$, and $Ea \leq \frac{3}{8}\theta^2$ follows from $4a(2a+b) \leq 3(a+b)^2$, equivalent to $(5a+3b)(a-b) \leq 0$. Thus $E \leq \frac{3}{8}m^{-1/2}$, which is at most $\frac{3}{8}(X-1)^{-1/2}$. Finally $(X-1)^{-1/2} \leq \frac{4}{3}X^{-1/2}$ for $n \geq 3$. □

Lemma 7.2 (gap and carry). For an integer $h \geq 1$, set $\Delta_h X(n) = X(n+2h) - X(n)$. Then

$$m(n+2h) - m(n) = \lfloor \Delta_h X(n) \rfloor + \kappa_h(n), \quad \kappa_h(n) = \mathbf{1}_{\{X(n)\}+\{\Delta_h X(n)\} \geq 1} \in \{0,1\}.$$

Proof. Expand $X(n) + \Delta_h X(n)$ into integer and fractional parts, and take its floor. The sum of the two fractional parts lies in $[0,2)$. $\square$

For $1 \leq h \leq P/4$ and $n \in (P, 2P]$, differentiation gives $(\Delta_h X)'(n) \asymp hP^{-1/2}$. The level sets of $\lfloor \Delta_h X \rfloor$ consequently form $O(1 + hP^{1/2})$ intervals. Full intervals have length $\asymp P^{1/2}/h$; the two endpoint intervals can be shorter. A coefficient involving this floor is frozen only on these intervals. A derivative estimate proved there must be summed over the partition, or replaced by a separate global argument with controlled errors.

Lemma 7.3 (next-level defect). One has

$$v^{3/2} = m^{9/4} - \tfrac{3}{2} v^{1/2}\theta_2 - \mathcal{R}, \qquad 0 \leq \mathcal{R} \leq \tfrac{3}{8} v^{-1/2}\theta_2^2.$$

Proof. Apply Taylor's theorem to $x^{3/2}$ between v and $v + \theta_2 = Y$. Its second derivative is $\frac{3}{4}x^{-1/2} \leq \frac{3}{4}v^{-1/2}$ on that interval, and is positive. Rearranging the Taylor formula gives the result. $\square$

These identities alone do not supply cancellation. The restricted estimate from the earlier draft,

$$\left| \sum_{\substack{P<n\leq 2P \\ n \text{ odd}}} e\big(\tfrac{i}{2} n^{3/2} + \tfrac{j}{2} m^{3/2}\big) \right| \ll_\varepsilon P^{23/24+\varepsilon}, \quad |i| \leq 2P^{1/24}, \quad 1 \leq |j| \leq 2P^{1/24}, \tag{7.1}$$

now follows from Theorem 4.5 with $k = 0$ and $C = 2$. The stronger general kernel target, with $c_k(n) = \frac{3k}{4} n^{9/8}$, is

$$K_k(P) = \sum_{\substack{P<n\leq 2P \\ n \text{ odd}}} e(c_k(n)\theta_2(n)), \qquad |K_k(P)| \ll_\varepsilon P^{95/96+\varepsilon}, \quad 1 \leq k \leq P^{1/24}. \tag{7.2}$$

The indices i, j, k are integers. Estimate (7.1) is unconditional in this revision; (7.2), with its printed uniformity, remains an **open target**. A bare-kernel bound does not imply a mixed-mode estimate by itself. The exact families needed here are instead proved in Theorems 4.9 and 4.11. Lemma 4.4 handles half-integer coefficients $j/2$ directly. This does not extend the frequency domain of the earlier general decorated lemma, whose separate applications remain unproved.

Short-interval variants require separate estimates. A bound for a transition set $\Omega \subset (P, 2P]$ of the form $|\Omega| \ll P^a$ implies only $|\Omega \cap I| \leq \min(|I|, O(P^a))$. It does not imply a bound proportional to $|I|/P$. Therefore the earlier localized-kernel threshold $|I| \geq P^{29/48+\delta}$ is not asserted here.

7.1 An unconditional estimate averaged over a shift

Proposition 7.4 (shift average). Let $L \geq 1$, let $A_1 < \cdots < A_L$ obey $|A_t - A_s| \geq a|t-s|$ with $a > 0$, and let $x_1, \ldots, x_L$ be real. For $S_\lambda = \sum_{t=1}^{L} e(A_t\{x_t + \lambda\})$,

$$\left| \int_0^1 |S_\lambda|^2 \, d\lambda - L \right| \leq \frac{4}{\pi} \frac{L}{a}(1 + \log L). \tag{7.3}$$

For every $0 < \eta < 1$, outside a set of shifts of measure at most η,

$$|S_\lambda| \leq \left[\frac{L}{\eta} \left(1 + \frac{4}{\pi a}(1 + \log L) \right) \right]^{1/2}.$$

Proof. Expanding the square gives the diagonal term L. For $t \neq s$, the integrand $e(A_t\{x_t + \lambda\} - A_s\{x_s+\lambda\})$ is periodic in λ. Integrate over a period starting at one of its two fractional-part breakpoints. There is at most one other interior breakpoint, giving at most two intervals on which the phase is affine with slope $A_t - A_s$. On each interval the integral has modulus at most $1/(\pi|A_t - A_s|)$. Hence the off-diagonal contribution is bounded by

$$\frac{2}{\pi a}\sum_{t\neq s}\frac{1}{|t-s|} = \frac{4}{\pi a}\sum_{r=1}^{L-1}\frac{L-r}{r} \leq \frac{4}{\pi}\frac{L}{a}(1+\log L).$$

For $L = 1$ the off-diagonal sum is empty. Markov's inequality applied to $|S_\lambda|^2$ gives the second assertion. □

The estimate allows a small exceptional set of shifts. It gives no bound for a specified shift, including $\lambda = 0$ in (7.2).

7.2 Counting the gap cells in a collision estimate

The following lemma repairs the cell-counting problem for a fixed curvature model. It does not assert that every decorated phase in the earlier kernel proof meets its hypotheses.

Lemma 7.5 (curvature estimates over a partition). Partition an interval of length at most P into interval cells, so that every subinterval of length L meets $O(1+\nu L)$ cells. On each cell let f be a real C^2 function; its values and derivatives may jump at cell boundaries. Suppose that a continuous reference function Λ satisfies $|\Lambda| \ll M$, and that, for $0 < s \leq 1$, both sets

$$\{|\Lambda| \leq sM\}, \qquad \{sM < |\Lambda| \leq 2sM\}$$

are unions of $O(1)$ intervals, each set of total length $O(Ps)$. Assume

$$|f'' - \Lambda| \leq \rho M \quad \text{on every cell}, \qquad 0 \leq \rho \leq 1/8, \quad 0 < M \leq 1, \quad MP^2 \geq 1.$$

All constants in these assumptions are fixed. Then, summing over integers or over odd integers with the same implicit-constant convention,

$$\Big|\sum e(f(n))\Big| \ll PM^{1/2} + (1+\nu P)M^{-1/2} + (P/M)^{1/3} + P\rho + 1. \tag{7.4}$$

Proof. Take $\tau = \max(4\rho, (MP^2)^{-1/3})$. If τ is bounded below by a positive absolute constant, the terms $P\rho + (P/M)^{1/3}$ already give the trivial bound $O(P)$. Otherwise discard $\{|\Lambda| \leq \tau M\}$, containing $O(P\tau+1)$ integers. Split the remainder into dyadic bands $|\Lambda| \asymp sM$, with $\tau \ll s \ll 1$, and the outer set $|\Lambda| > M$ if necessary. Each band has total length $O(Ps)$ and meets $O(1+\nu Ps)$ cells. On its intersection with a cell the curvature has one sign and size $\asymp sM$.

Apply the second-derivative estimate on every such intersection. The resulting costs at scale s are

$$O\big(PM^{1/2}s^{3/2} + M^{-1/2}s^{-1/2} + \nu PM^{-1/2}s^{1/2}\big).$$

The geometric sum over bands is $O(PM^{1/2} + M^{-1/2}\tau^{-1/2} + \nu PM^{-1/2})$. Our choice gives $M^{-1/2}\tau^{-1/2} \leq (P/M)^{1/3}$ and $P\tau \ll P\rho + (P/M)^{1/3}$, proving (7.4). One may assign boundary integers to either adjacent cell and estimate that cell up to its endpoint. □

Proposition 7.6 (the basic frozen-gap collision model). Let $u \ge 1/2$, $1 \le h \le P^{1/8}$, $uh \le P^{1/2}$, and let $w > 0$ satisfy $c_1 uhP^{-1/4} \le w \le c_2 uhP^{-1/4}$, for fixed $0 < c_1 < c_2$. With A_h as in Lemma 4.4 and $G(x) = \lfloor (x+2h)^{3/2} - x^{3/2} \rfloor$, put

$$f(x) = uA_h(x) + \tfrac{3u}{2} G(x)(x+2h)^{3/4} + wx^{3/2}.$$

Then

$$\left| \sum_{\substack{P<n\le 2P\\ n \text{ odd}}} e(f(n)) \right| \ll_{c_1,c_2} (uh)^{1/2}P^{5/8} + (h/u)^{1/2}P^{7/8} + (uh)^{-1/3}P^{7/12} + h + P^{1/2}/h + 1. \quad (7.5)$$

Proof. On a gap cell the frozen-coefficient derivative is

$$f''(x) = -\tfrac{9}{32} uG(x)(x+2h)^{-5/4} + \tfrac{3}{4} wx^{-1/2} + uA_h''(x).$$

Use the reference function

$$\Lambda(x) = -\tfrac{27}{32} uhx^{-3/4} + \tfrac{3}{4} wx^{-1/2}, \qquad M = uhP^{-3/4}.$$

The identities $G(x) = 3h\sqrt{x} + O(1 + h^2P^{-1/2})$ and $A_h'' = O(h^2P^{-7/4})$ show, as a comparison of values,

$$|f'' - \Lambda| \ll M\big(h/P + (hP^{1/2})^{-1}\big).$$

This does not differentiate G. The gap-cell partition has density parameter $\nu \asymp hP^{-1/2}$.

To verify the sublevel conditions, factor $\Lambda(x) = \frac{3}{4}x^{-3/4}(wx^{1/4} - 9uh/8)$. The expression in parentheses is strictly increasing, with derivative $\asymp uh/P$ in the stated collision range. Hence $|\Lambda| \le sM$ occupies length $O(Ps)$. Moreover Λ' has at most one zero, so the sublevel sets and dyadic bands have a bounded number of components. Apply Lemma 7.5 with $\rho = O(h/P + (hP^{1/2})^{-1}) = o(1)$. Its terms give (7.5); the extra $M^{-1/2}$ is absorbed by $(h/u)^{1/2}P^{7/8}$. □

For integer w in the collision range, $w \ge 1$ implies $uh \gg P^{1/4}$. Formula (7.5) is then $O(P^{7/8})$: its second term is $h(uh)^{-1/2}P^{7/8} \ll P^{7/8}$, and the remaining terms are no larger. This verifies that the basic integer collision band leaves a power saving.

Thus the transition cost need not be paid independently on every gap cell: a global sublevel argument counts how many cells can meet each curvature band. To use this in the complete kernel proof one must still verify the reference-function hypotheses, partition density, and weighted sums for all additional Fourier modes and decorations. Neither (7.4) nor (7.5) supplies that verification, and neither rescales a transition-set bound by an arbitrary short-interval length.

8. Evidence, availability, and remaining work

The status of the principal statements is as follows.

Statement	Warrant in this version
Envelope, branch identity, minimal words, floor defects	Exact proofs in Sections 2, 5, and 7
Certificate density 3/4	Classical analytic proof in Theorem 3.1
Correlation-to-count and Fourier-to-parity implications	Proofs in Propositions 4.1 and 4.2
Certificate density 13/16	Unconditional: Corollary 4.6 and Theorem 5.2
Certificate subfamily density 27/32	Unconditional: Corollary 4.10 and Theorem 5.3
Full five-step certificate density 7/8	Unconditional: Corollary 4.12, Theorem 5.4, and Appendices A–C

Statement	Warrant in this version
Density-one finite certificates	Conditional on FD; no depth uniformity assumed
Restricted mixed sums	Theorems 4.5, 4.9, and 4.11
General decorated kernel and short-interval extension	Unproved; no numerical threshold certified
Basic frozen-gap collision model	Proposition 7.6; does not certify every decorated phase
Shift-averaged estimate	Direct integral proof in Proposition 7.4

Supporting software is available in the Balanced Ternary Mathematical Laboratory repository, `https://github.com/sneakyweasel/btlab`. The accompanying source package contains this manuscript, its LaTeX build, and an exact-arithmetic validation script. That script checks the finite certificate list, the indicator identity on a finite census, and selected algebraic inequalities. A second exact-arithmetic script checks the differenced identity, carry expansion, and exponent comparisons used in the four-step repairs. Further exact scripts check the fifth-letter centering identity, the D2 floor reduction, the signed-wave and master inventories, the mixed curvature coefficients, and all displayed exponent budgets. The source package includes these scripts and their results. Those checks support reproducibility; they are not proofs of asymptotic cancellation.

Earlier repository audits and Lean modules concern identities, inequalities, and arithmetic from the previous draft. This version claims no Lean formalization of its analytic hypotheses or of the complete manuscript. Earlier theorem numbers are superseded and should not be used as citations to unconditional nested estimates. Companion texts that rely on those estimates require their own review.

The full five-step certificate class proved here has density 7/8. The argument does not extend its error estimates proportionally to an arbitrary short interval, and it supplies no universal termination conclusion. A proof of all fixed-depth fair-share counts would imply density-one certificates by Theorem 6.1; that additional hypothesis remains open.

Appendix A. Floor reductions and the carry inventory

We collect the estimates used by the OOOEE argument. All counts of exceptional integers use the original gap runs, before any later frequency partition. Constants may depend on fixed comparability constants. Every displayed estimate is uniform for sufficiently large P; bounded smaller P is absorbed into the implied constants.

A.1 A reduction for a differenced floor factor

Lemma A.1 (fixed-label floor reduction). The definitions below lead to (A.9), with the stated error and coefficient variation bounds.

In Appendices A–C, a subscript on Δ denotes the actual translation: $\Delta_d f(x) = f(x+d) - f(x)$. Thus a half-shift h uses Δ_{2h}. This differs from the local half-shift convention $\Delta_h f(x) = f(x+2h) - f(x)$ in Section 4.2. All constants below are absolute, or depend on fixed constants in comparability bounds. Let P tend to infinity, with integer parameters

$$1 \le k, h_1, h_2 \le P^{1/24}, \qquad 1 \le h \le P^{1/8}.$$

Put X(x)= $x^{3/2}$, m(n)=floor(X(n)), and c(x)=3k $x^{9/8}$ /4. Work on (P,2P], extending the definitions to (P,3P] to accommodate shifts. Fixed positive multiples of the upper parameter bounds are allowed, with constants depending on those multiples. All estimates also hold after restriction to odd integers.

Partition by the integer parts of

$$X(x+2h_1)-X(x), \quad X(x+2h_2)-X(x), \quad X(x+2h_1+2h_2)-X(x).$$

There are

$$D \ll (h_1+h_2)P^{1/2}$$

runs, and an interval of length L meets $O(1+(h_1+h_2)LP^{-1/2})$ runs.

On each run freeze a carry label and its integer offsets β_1, β_2, β_{12}. Assume β_i is comparable to h_i $P^{1/2}$, for i=1,2, and j= β_{12} - β_1 - β_2 is a bounded integer, with |j| $\leq$ 2 for the admissible branches used here. Define, on that run,

$$\begin{aligned} F(t) &= (t+\beta_1+\beta_2+j)^{3/2}-(t+\beta_1)^{3/2}-(t+\beta_2)^{3/2}+t^{3/2}, \\ G(x) &= F(X(x)), \qquad A(n) = \lfloor F(m(n)) \rfloor. \end{aligned}$$

The offsets may change between runs. The theorem treats any such piecewise extension; no derivative crosses a run boundary. There are only finitely many carry labels, so bounds for their positive error terms can be summed. Reassembling oscillatory sums with the actual carry indicators is a further step, not an assumption that a branch extension is the physical orbit everywhere.

On an interval I of length L, suppose G is C^1 and a $\leq$ |G'| $\leq$ C a, with 0<a $\leq$ 1. Its derivative has constant sign. For 0< $\delta \leq 1/2$, the number of integer n in I with distance(G(n),Z) $\leq \delta$ is

$$\ll \delta(L+a^{-1})+aL+1. \tag{A.1}$$

Indeed G(I) meets $O(aL+1)$ integer neighborhoods. Each inverse image is an interval of length $O(\delta/a)$, containing $O(\delta/a+1)$ integers. This includes endpoints and exact integer values.

Let

$$E_Q(t)=\min(1,(Q\|t\|)^{-1}), \qquad E_Q(t)=1 \quad (t\in\mathbb{Z}).$$

Splitting the distances to Z into dyadic bands gives

$$\sum_{n\in I} E_Q(G(n)) \ll \frac{L+a^{-1}}{Q}\log(2Q)+aL+1. \tag{A.2}$$

The additive lattice-count cost is multiplied by a convergent geometric sum, not by the number of bands.

Over D disjoint runs with a common comparable derivative scale, these become

$$\begin{aligned} \#\{\|G(n)\|\leq\delta\} &\ll \delta(P+D/a)+Pa+D, \\ \sum E_Q(G(n)) &\ll \frac{(P+D/a)\log(2Q)}{Q}+Pa+D. \end{aligned} \tag{A.3}$$

Runs of different offset types are bounded separately. No equidistribution assumption, random model, or rescaling of a global exceptional set is used.

Set p= h_1 h_2. The zero-offset part has the exact integral representation

$$F_0(t)=\frac{3}{4}\int_0^{\beta_1}\int_0^{\beta_2}(t+s+r)^{-1/2}\,dr\,ds.$$

Consequently, with the β values fixed when differentiating,

$$\begin{aligned} G_0'(x) &= -\frac{9}{16}\beta_1\beta_2 x^{-7/4}(1+o(1)), \\ G_0''(x) &= \frac{63}{64}\beta_1\beta_2 x^{-11/4}(1+o(1)). \end{aligned}$$

The errors are uniform because (β_1 + β_2)/X=o(1).

The offset contribution is

$$\frac{3j}{2}\int_0^1 (t+\beta_1+\beta_2+sj)^{1/2}\,ds.$$

After composition with X its first derivative is (9j/8) $x^{-1/4}$ (1+o(1)). For j nonzero it dominates the zero-offset derivative, since p $P^{-1/2}$ =o(1). Thus, on every run,

$$|G'| \asymp \begin{cases} pP^{-3/4}, & j=0, \\ P^{-1/4}, & j\neq 0, \end{cases} \qquad |G''| \ll |j|P^{-5/4} + pP^{-7/4}. \tag{A.4}$$

Also |G| is $O(|j|P^{3/4} + pP^{1/4})$. These are derivatives of frozen-offset functions, not derivatives of the approximations β_i approximately 3h_i $\sqrt{x}$.

Take Q=floor($P^{5/16}$). For j=0, the potentially largest term in (A.3) is

$$\frac{D}{Qa} \ll \frac{h_1+h_2}{h_1h_2}P^{15/16} \ll P^{15/16}.$$

The other terms are $P^{11/16}$, $pP^{1/4}$, and D. For j nonzero, (A.3) is $O(P^{3/4}\log P)$. It follows that, summed over all runs of either type,

$$\sum E_Q(G(n)) \ll P^{15/16}\log P. \tag{A.5}$$

The replacement of m by X is used only inside the floor, after counting its exceptional set. The mean value theorem gives

$$|F(m(n)) - G(n)| \ll |j|P^{-3/4} + pP^{-5/4} =: \delta.$$

A floor mismatch requires G(n) to lie within δ of an integer. For either derivative scale in (A.4), δ /a= $O(P^{-1/2})$. The first line of (A.3) therefore proves

$$\#\{n : A(n) \neq \lfloor G(n) \rfloor\} \ll P^{3/4}. \tag{A.6}$$

This remains useful when c is large: changing a unit exponential costs at most two on the exceptional set. A large coefficient is never multiplied by an uncontrolled pointwise floor error.

Exclude for the moment n whose interval from n to n+2h crosses an original run boundary. There are $O(hD)$ such integers.

Within a run, G is monotone. If floor(G(n+2h)) differs from floor(G(n)), this interval contains an integer-level crossing of G. There are $O(aL+1)$ levels on a run of length L, and any crossing belongs to at most $O(h)$ such integer starting intervals. Including the original run boundaries gives

$$\#\{\text{floor change or run crossing}\} \ll h(Pa+D) \ll P^{7/8}. \tag{A.7}$$

For zero offset the bound is even smaller; the uniform largest term is $hP^{3/4}$ from nonzero offsets.

Write B(x)=c(x+2h)-c(x). Using (A.6) at both endpoints and then (A.7), we obtain the following statement in the sum of absolute errors:

$$e\big(-\Delta_{2h}(cA)(n)\big) = e\big(-B(n)\lfloor G(n)\rfloor\big) + \mathcal{E}_n, \qquad \sum |\mathcal{E}_n| \ll P^{7/8}. \tag{A.8}$$

Both expressions in this comparison are unimodular. The second expression is defined on every run, including the exceptional starting points; no irregularly punctured domain is passed to a derivative test.

The floor-crossing argument also explains why a small drift is insufficient by itself. For comparison, on zero-offset runs |Delta G| is $O(hpP^{-3/4})$; at the carry cutoff J=floor($P^{1/8}$), J|Delta G|=o(1). Therefore the distance majorant E_J (Delta G) equals one for sufficiently large P. The standard sawtooth Fourier approximation has an order-one endpoint error there as well. Small drift alone cannot justify the $O(P/J)$ charge for this individual layer. The exact carry combination may cancel; (A.7) preserves that cancellation by counting the floor change before expansion.

The coefficient B is increasing, with

$$B \asymp khP^{1/8}, \quad B' \asymp khP^{-7/8}, \quad |B''| \ll khP^{-15/8}.$$

Refine the original runs by N=floor(B), and put β =B-N. The extra number of cuts is $O(1 + khP^{1/8})$. The exact identity

$$e(B\{G\}) = e(NG)e(\beta\{G\})$$

centers the frequency before truncation.

For $0 \le \beta \le 1$ let

$$a_r(\beta) = \int_0^1 e((\beta - r)t)\, dt.$$

By Lemma 4.7, or its complex conjugate,

$$e(\beta\{G\}) = \sum_{|r|\le Q} a_r(\beta)e(rG) + O(E_Q(G)).$$

The integral definition includes the removable singularity. Both | a_r | and | a_r '| are $O(1/(1+|r|))$. On each refined interval, β is monotone and has variation at most one, so

$$\sum_{|r|\le Q} (\|a_r(\beta)\|_\infty + \mathrm{TV}(a_r(\beta))) \ll \log(2Q).$$

Combining this expansion with (A.5) and (A.8) proves the fixed-label floor reduction:

$$\begin{aligned} e\big(-\Delta_{2h}(cA)(n)\big) &= \sum_{|r|\le Q} a_r(\beta(n))e(\Phi_r(n)) + \mathcal{R}_n, \\ \Phi_r(x) &= (N + r - B(x))G(x), \\ \sum_{P<n\le 2P} &|\mathcal{R}_n| \ll P^{15/16}\log P. \end{aligned} \tag{A.9}$$

N and all offsets are fixed only within the refined interval. The error sum is charged once on the original runs; it is not multiplied by the number of frequency windows. This is valid with any additional weight of modulus at most one. A fixed number of such floor factors can be expanded by telescoping, at a fixed extra power of log P and with the same power saving.

This replacement uses the bounded residual β, so it has no unaccounted continuous-part tail between Q and $P^{1/2}$.

Differentiate with N and r fixed:

$$\Phi_r'' = (N + r - B)G'' - 2B'G' - B''G.$$

Since N+r-B=r- β, (A.4) yields

$$|\Phi_r''| \ll (Q + 1)(|j|P^{-5/4} + pP^{-7/4}) + kh(|j|P^{-9/8} + pP^{-13/8}). \tag{A.10}$$

For an undisturbed wave curvature $M = uhP^{-3/4}$, u $\ge 1/2$, the four exponent bounds for this ratio are respectively

$$P^{-3/16}, \quad P^{-29/48}, \quad P^{-1/3}, \quad P^{-3/4}.$$

Thus $|\Phi_r''|= O(P^{-3/16}M)$, uniformly.

Where a total phase already has curvature comparable to M, the extra endpoint cost in the second-derivative estimate is

$$
\begin{aligned}
(D+1+khP^{1/8})M^{-1/2} \ll (h_1+h_2)(uh)^{-1/2}P^{7/8} \\
+ k(h/u)^{1/2}P^{1/2} + (uh)^{-1/2}P^{3/8}.
\end{aligned}
\tag{A.11}
$$

Its largest uniform exponent is 11/12, below 15/16. Partial summation uses the coefficient-variation bound proved above.

This comparison is not a declaration that an arbitrary sum of signed waves has curvature comparable to M. At a curvature cancellation, the reference-function and sublevel hypotheses must still be verified for the complete phase.

A.2 Joint first-floor carries

Lemma A.2 (carry arcs and global positive errors). Let $1 \leq h \leq P^{1/12}$, and let the number of shifts below be bounded by a fixed constant.

There are finitely many first-floor evaluations in the main wave and the D1 terms. All have the form m(n+s), with s an even nonnegative integer $O_{C,M}(h+P^{1/24})$. Use the single variable θ ={X(n)} and

$$
m(n+s)-m(n) = b_s + \mathbf{1}_{\theta\geq 1-\delta_s}, \qquad b_s = \lfloor X(n+s)-X(n)\rfloor, \quad \delta_s = \{X(n+s)-X(n)\}.
$$

The convention when δ_s =0 is that this indicator is zero for $0 \leq \theta$ <1.

Partition by the integer gaps b_s and any input or coefficient windows already required by the argument. There are $O(1+\sum_s s\sqrt{P})$ original gap runs. Their local count in an interval of length L is $O(1+\sum_s sLP^{-1/2})$. On each gap run each δ_s is monotone and varies by at most one. Sort their endpoints 1- δ_s; two distinct endpoints cross at most once on such a run, because the difference of the corresponding real gaps is monotone. Equal shifts give identical endpoints. This adds only $O_{C,M}(1)$ cuts per cell.

A joint carry pattern is now a fixed finite union of intervals of θ, with endpoints 0,1, or 1- δ_s. Its Fourier expansion truncated at R=floor($P^{1/4}$) has: - a zero-mode coefficient with sup norm plus variation $O_{C,M}(1)$; - endpoint modes with coefficients $O_{C,M}(1/|v|)$, 0<|v| $\leq$ R, multiplying e(vX(n+s)) for one of the allowed s (s=0 included); - an absolute remainder bounded by a constant times the sum of E_R (X(n+s)) over these finitely many endpoints, where E_R (z)=min(1,1/(R||z||)), with value one at integers.

This is the usual interval Fourier expansion: a moving endpoint u=1- δ_s supplies e(-v u), and, since b_s is integer, e(vX(n))e(-v(1- δ_s))=e(vX(n+s)). Thus the moving endpoint belongs in the phase, not in an unbounded-variation coefficient. Zero-length intervals and points at endpoints are covered by the positive remainder.

Here is the global bound needed for that remainder:

$$
\sum_{P<n\leq 2P} E_R(X(n+s)) \ll_{C,M} P^{5/6}\log(2P). \tag{A.12}
$$

It holds also over odd integers or a subinterval. To verify it, the second-derivative test for vX(x+s) gives $O(|v|^{1/2}P^{3/4}+|v|^{-1/2}P^{1/4})$. The interval-discrepancy inequality with auxiliary cutoff K=floor($P^{1/6}$) then gives discrepancy $O(P^{5/6})$. Dyadic bands for the distance to an integer give $O(P\log(2R)/R+P^{5/6})$, which implies (A.12). The remainder is charged once over the common input interval. It is not multiplied by the cell count. Products with the other finite Fourier expansions introduce at most a fixed logarithmic factor.

A.3 Positive errors for the second-level inventory

Let $X = x^{3/2}$, $m = \lfloor X \rfloor$, $Y = m^{3/2}$, and put $d_i = 2h_i$, $W_i = \Delta_i Y$, $D = \Delta_1\Delta_2 Y$, where $1 \le h_1 \le P^{1/48}$, $1 \le h_2 \le P^{1/24}$, and $p = h_1 h_2$. Here $\Delta_i f(x) = f(x + d_i) - f(x)$. Even translations by $O(P^{1/24})$ are allowed. **Lemma A.3 (positive inventory errors).**

Set $J = \lfloor P^{1/24} \rfloor$, and use $E_J(z) = \min(1, (J\|z\|)^{-1})$, with value one at integers. For every argument occurring in the carry expansion below,

$$\sum_{P<n\le 2P,\ n \text{ odd}} E_J(Z(n)) \ll P^{23/24}(\log(2P))^C, \tag{A.13}$$

where

$$Z \in \{Y(n), Y(n+d_1), Y(n+d_2), Y(n+d_1+d_2), W_1(n), W_2(n), W_1(n+d_2), D(n)\}.$$

The constant C is absolute. This is an unweighted positive bound; it does not assume that an anchor is still present.

For Y at any of these even translations, Theorem 4.5, with modes $(i, j, k) = (0, 2q, 0)$, gives $O(P^{23/24})$ for $1 \le$ |q| $\le$ J. The discrepancy inequality at cutoff J, followed by dyadic distance bands, proves (A.13). The same argument for W uses Lemma 4.4 with i=k=0 and j=2q. Its bound is $O(P^{7/8}(1 + h_i^{1/2})) = O(P^{43/48})$, which also fits (A.13). The proofs of those results work on any subinterval of [P,3P]: their length, gap-cell, and positive-error bounds are all in the ambient scale P. No proportional rescaling of an exceptional set is used for the translated intervals here.

For D, a separate elementary argument is needed. Put

$$j(n) = \Delta_1\Delta_2 m(n), \qquad -1 \le j(n) \le 2.$$

The exact frozen-branch integral formulas imply the global value approximation

$$D(n) = g_{j(n)}(n) + O((h_1+h_2)P^{-1/4}), \qquad g_j(x) = \tfrac{3}{2} j x^{3/4} + \tfrac{27}{4} p x^{1/4}. \tag{A.14}$$

To verify the error, the first-floor offsets satisfy β_i =3h_i $\sqrt{n}$ + $O(1)$, hence β_1 β_2 =9pn+ $O((h_1+h_2)\sqrt{P}+1)$. In the zero-offset integral, replacing its argument by $n^{3/2}$ has relative error $O((h_1+h_2)/P)$; in the offset integral the corresponding absolute error is $O((h_1+h_2)P^{-1/4})$. The resulting extra zero-offset error $O(p(h_1+h_2)P^{-3/4})$ is smaller than the displayed error. This is a comparison of values, not of frozen derivatives.

The error in (A.14) is $O(P^{-5/24})$, so J times that error tends to zero. For |u-v| $\le$ 1/(2J), one has E_J (u) $\le$ 2E_J(v). On a branch j=0, g_j has derivative comparable to $pP^{-3/4}$. On a nonzero branch it has derivative comparable to $P^{-1/4}$, of the sign of j. Both are monotone slow variables on the whole dyadic block. The monotone counting lemma (A.2) in Appendix A.1 gives

$$\sum E_J(g_j(n)) \ll (P + a_j^{-1})\log(2J)/J + Pa_j + 1, \quad a_0 \asymp pP^{-3/4}, \quad a_j \asymp P^{-1/4}\ (j \ne 0).$$

Every term is $O(P^{23/24}\log P)$. Bound a branch-restricted positive sum by this full-block sum and add the four possible j values. This proves (A.13) for D without multiplying a floor-error estimate by the number of original gap runs.

A.4 Exact centered master inventory

Let $c(x) = 3kx^{9/8}/4$, $c_{11}(x) = c(x + d_1 + d_2)$, $\vartheta = \{Y\}$, and $1 \le k \le CP^{1/24}$. **Lemma A.4 (master identity and coefficient mass).**

Define the binary carry $C(A, B) = \{A\} + \{B\} - \{A + B\}$, and set

$$\kappa_1 = C(Y, W_1), \quad \kappa_2 = C(Y, W_2), \quad \kappa_1^+ = C(Y(n+d_2), W_1(n+d_2)), \quad \kappa_* = C(W_1, D).$$

Let

$$B_2(x) = \Delta_2 c(x + d_1), \qquad B_1(x) = \Delta_1 c(x + d_2), \qquad N_i(x) = \lfloor B_i(x) \rfloor.$$

The product rule and the exact carry identity give

$$\begin{aligned} \Delta_1\Delta_2(c\vartheta) = (\Delta_1\Delta_2 c)\vartheta + B_2(\{W_1\} - \kappa_1) + B_1(\{W_2\} - \kappa_2) \\ + c_{11}(\{D\} - \kappa_* - \kappa_1^+ + \kappa_1). \end{aligned} \tag{A.15}$$

With $\Pi = kh_1h_2$, the first term has total deletion cost $O(\Pi P^{1/8}) = O(P^{11/48})$.

For each factor e(B_i {W}), center at the integer N_i:

$$e(B_i\{W\}) = e(N_iW)e(\beta_i\{W\}), \qquad \beta_i = B_i - N_i \in [0, 1).$$

Here β_i denotes a coefficient residual only; it is not a first-floor offset used in a branch formula. Expand the last factor at the short cutoff J:

$$e(\beta_i\{W\}) = \sum_{|r|\le J} a_r(\beta_i)e(rW) + O(E_J(W)), \quad a_r(\beta) = \int_0^1 e((\beta - r)y)\, dy. \tag{A.16}$$

Its coefficients satisfy | a_r |+| a_r '|«1/(1+|r|), including removable singularities. The error is covered by (A.13), independently of the large center N_i. There is no uncentered large-coefficient tail to discard.

For each of the five carry exponentials in (A.15), use exactly

$$e(g\kappa) = 1 + \kappa(e(g) - 1).$$

Expand the factor e(g)-1 into two terms and move e(g) into the smooth phase. Expand kappa additively using its three unit fractional parts, each at cutoff J by Lemma 4.3. The only arguments are those in (A.13), because

$$Y + W_i = Y(n + d_i), \quad W_1 + D = W_1(n + d_2), \quad Y(n + d_2) + W_1(n + d_2) = Y(n + d_1 + d_2).$$

The smooth phases introduced this way are bounded integer combinations of B_1, B_2, c_{11}.

There are two centered Fourier layers and at most five carry layers. Their total coefficient mass is $O((\log(2P))^7)$; a larger fixed logarithmic power covers all errors. Indeed, pointwise truncated factors have $O(\log P)$ bounds, so telescoping products multiplies each error in (A.13) only by a fixed logarithmic power.

The N_i windows are retained globally as functions of x. Both B_i are increasing, with

$$B_i \asymp kh_iP^{1/8}, \quad B_i' \asymp kh_iP^{-7/8}.$$

Their common partition has local count $O(1 + k(h_1 + h_2)LP^{-7/8})$ and total count $O(P^{5/24})$. On each cell the residuals β_i are monotone in [0,1), so the same logarithmic bound holds for summed coefficient sup norms plus variations.

For arbitrary four corner coefficients, the exact identity is

$$\sum_{d\in\{0,d_1,d_2,d_1+d_2\}} q_dY(n + d) = tY + (q_{d_1} + q_{d_1+d_2})W_1 + (q_{d_2} + q_{d_1+d_2})W_2 + q_{d_1+d_2}D, \quad t = \sum_d q_d. \tag{A.17}$$

Use also W_1 (n+ d_2)= W_1 +D.

Consequently (A.15)–(A.16) reduce the sum with phase $Delta_1$ $Delta_2$ (c ϑ), with absolute error $O(P^{23/24}\log^C(P))$, to a finite weighted family with phases

$$c_{11}\{D\} + tY + (N_2 + u)W_1 + (N_1 + v)W_2 + qD + \phi. \tag{A.18}$$

Here t,u,v,q are fixed integers for each Fourier multi-index,

$$|t|, |u|, |v|, |q| \leq 6J,$$

and ϕ is a bounded integer combination of B_1, B_2, c_{11}. In particular

$$|\phi''| \ll kP^{-7/8}, \qquad |\phi'''| \ll kP^{-15/8}. \tag{A.19}$$

The two centered residual frequencies account for the extra unit in the bound 6J; each carry layer contributes at most one mode. All signs and zero coordinates are included.

More precisely there are nonnegative numbers ρ_ν, one for each multi-index, with sum $\rho_\nu = O(\log^7(P))$, such that its amplitude divided by ρ_ν has bounded sup norm plus variation on each N_1, N_2 cell. This follows by multiplying the harmonic coefficient bounds above and applying the product variation inequality. Thus it suffices to prove a uniform bound for each normalized amplitude; no estimate is multiplied by the number of frequency windows.

The first-difference coefficients satisfy

$$|(N_2 + u)h_1| + |(N_1 + v)h_2| \ll \Pi P^{1/8} + J(h_1 + h_2) \ll P^{11/48}, \tag{A.20}$$

well within the widened $P^{1/2}$ budget.

Appendix B. A wave-bearing estimate

The following estimate is used only for the family that its statement specifies. In Appendix C the total Y frequency is a nonzero integer. Allowing half-integer frequencies here also covers the elementary main-wave argument.

Theorem B.1 (nonzero total frequency).

Let $e(v) = \exp(2\pi i v)$, and put

$$X(x) = x^{3/2}, \quad m(n) = \lfloor X(n) \rfloor, \quad Y(n) = m(n)^{3/2}, \quad \Delta_d f(x) = f(x+d) - f(x).$$

All functions are defined on [P,3P], with harmless further extension at its upper endpoint if needed. Constants C and M are fixed. All estimates concern sufficiently large P depending only on them. Let t be real and fixed throughout the sum, with

$$\tfrac{1}{2} \leq |t| \leq CP^{1/24}.$$

There are at most M first-difference terms, with even shifts $e_j \geq 0$ and positive integer half-shifts r_j satisfying

$$e_j \leq CP^{1/24}, \qquad 1 \leq r_j \leq CP^{1/24}, \qquad |a_j(x)| r_j \leq CP^{1/2}. \tag{B.1}$$

The real coefficients a_j are constant separately on the cells of an interval partition $\mathcal{P}$. Any interval of length L meets at most C(1+L $P^{-11/24}$) cells of this partition. In particular, its total cell count is $O_C(P^{13/24})$. The real function ϕ is C^3 on each cell and satisfies

$$|\phi'''(x)| \leq CP^{-13/12}. \tag{B.2}$$

No continuity across cell boundaries is required. The complex amplitude η satisfies

$$\|\eta\|_{\infty,I} + \mathrm{TV}_I(\eta) \leq C$$

on each cell I. Endpoint values are assigned consistently; their possible exceptional contribution is covered by the cell count.

We also allow at most M terms $-\sigma_\ell c_\ell A_\ell$, where σ belongs to {-1,0,1}. These are the fixed-label D2 objects of the Appendix A.1, with

$$1 \leq k_\ell, h_{\ell,1}, h_{\ell,2} \leq CP^{1/24}, \qquad c_\ell(x) = \tfrac{3}{4} k_\ell (x + v_\ell)^{9/8}.$$

The integers k and h are positive. The base shift s_ℓ and the coefficient shift v_ℓ are nonnegative even integers $O_C(P^{1/24})$. To specify the object unambiguously, partition by the integer parts of

$$X(x+s_\ell+2h_{\ell,i})-X(x+s_\ell), \qquad X(x+s_\ell+2h_{\ell,1}+2h_{\ell,2})-X(x+s_\ell).$$

On every such original run choose fixed admissible carry offsets β_1, β_2, β_{12}, with β_i comparable to $h_i\sqrt{P}$ and j= β_{12} - β_1 - β_2 a bounded integer (|j| $\le$ 2 suffices). Set

$$\begin{aligned} F_\ell(z) &= (z+\beta_1+\beta_2+j)^{3/2}-(z+\beta_1)^{3/2}-(z+\beta_2)^{3/2}+z^{3/2},\\ G_\ell(x) &= F_\ell(X(x+s_\ell)),\\ A_\ell(n) &= \lfloor F_\ell(m(n+s_\ell))\rfloor. \end{aligned}$$

The version $A_\ell(n)=\lfloor G_\ell(n)\rfloor$ is also allowed.

The partition $\mathcal{P}$ refines these original runs. Further refinement does not change their chosen labels: it cannot introduce new arbitrary jumps in F_ℓ or G_ℓ. All D2 counting errors below are evaluated on the original runs. These floors are not first partitioned into their integer level sets.

Under these hypotheses the theorem is

$$\begin{gathered} U(P)=\sum_{\substack{P<n\le 2P\\ n \text{ odd}}}\eta(n)e\Big(tY(n)+\sum_j a_j(n)\Delta_{2r_j}Y(n+e_j)-\sum_\ell \sigma_\ell c_\ell(n)A_\ell(n)+\phi(n)\Big),\\ |U(P)|\ll_{C,M,\varepsilon} P^{31/32+\varepsilon}. \end{gathered} \tag{B.3}$$

The hypotheses on the partition and ϕ must be verified in any application; they do not permit arbitrary additional floor factors or a smooth twist that cancels the main wave.

B.1 The widened D1 linearization

Fix an A-process half-shift $1\le h\le P^{1/12}$. On an actual carry branch for z=n+e, put

$$m(z+2h)=m(z)+\beta_1,\quad m(z+2r)=m(z)+\beta_2,\quad m(z+2h+2r)=m(z)+\beta_1+\beta_2+j.$$

Then

$$\Delta_{2h}\Delta_{2r}Y(z)=F_j(m(z)),$$

where F_j has the same four-term form as F_ℓ above. Uniformly, β_1 is comparable to h $\sqrt{P}$, β_2 to r $\sqrt{P}$, and |j| $\le$ 2. The latter follows because the corresponding real mixed difference of X is $O(hrP^{-1/2})$ =o(1).

The exact integral formulas, with the β values frozen, are

$$\begin{aligned} F_0(v) &= \tfrac{3}{4}\int_0^{\beta_1}\int_0^{\beta_2}(v+b+c)^{-1/2}\,dc\,db,\\ F_j(v)-F_0(v) &= \tfrac{3j}{2}\int_0^1(v+\beta_1+\beta_2+\tau j)^{1/2}\,d\tau. \end{aligned} \tag{B.4}$$

They hold for both signs of j. Differentiating with respect to v gives

$$|F_j'(v)|\ll |j|P^{-3/4}+hrP^{-5/4}, \qquad v\asymp P^{3/2}. \tag{B.5}$$

Since 0 $\le$ X(z)-m(z)<1, replacing F_j (m(z)) by F_j (X(z)) in the exponential costs at most a constant times |a| times (B.5). The error summed over all integers in the dyadic interval is

$$\ll |a|P^{1/4}+|a|hrP^{-1/4}\ll_C P^{3/4}/r+hP^{1/4}\ll_C P^{3/4}. \tag{B.6}$$

This is one global charge; it is not paid again on each branch.

A second differentiation after composition gives

$$|(F_j \circ X)''| \ll |j| P^{-5/4} + hrP^{-7/4}.$$

Consequently the retained D1 curvature is at most

$$C\big(P^{-3/4}/r + hP^{-5/4}\big). \tag{B.7}$$

The leading frozen coefficients provide a useful check:

	$F_j'(v)$	$(F_j \circ X)''(x)$
offset part	$(3j/4)v^{-1/2}$	$-(9j/32)x^{-5/4}$
zero-offset part	$-(3\beta_1\beta_2/8)v^{-3/2}$	$(63\beta_1\beta_2/64)x^{-11/4}$.

These are leading terms with uniform smaller remainders, not exact equalities for finite β. One must not differentiate the moving approximation β_i approximately 3h_i $\sqrt{x}$.

No large Fourier expansion of this D1 floor error is needed.

B.2 Differencing and the nonzero main wave

Choose $H = \lfloor P^{1/12} \rfloor$. Apply van der Corput differencing to the sequence indexed by the odd integers, using shifts 2h:

$$|U|^2 \ll_C P^2/H + (P/H) \sum_{1 \le h < H} |V_h|. \tag{B.8}$$

Here V_h is the correlation on the overlap of the original interval and its translate. At most $O_C(hP^{13/24})$ starting points cross a cell boundary of $\mathcal{P}$. Their contribution is $O_C(P^{5/8})$ for each h. On each remaining stable interval I intersect (I-2h), all a_j are constant, the product of the two amplitudes has bounded variation $O_C(1)$, and

$$|(\Delta_{2h}\phi)''| \le 2h \sup_I |\phi'''| \ll_C hP^{-13/12}. \tag{B.9}$$

We apply all derivative tests to these intervals, never to a domain punctured by individual exceptional integers.

Let θ (n)={X(n)}. Taylor's theorem gives

$$Y(n) = n^{9/4} - \tfrac{3}{2}\theta(n)n^{3/4} + E(n), \qquad |E(n)| \ll P^{-3/4}.$$

On a fixed main carry branch write β =m(n+2h)-m(n), and put

$$A_h(x) = \Delta_{2h}(x^{9/4}) - \tfrac{3}{2}\Delta_{2h}X(x)(x+2h)^{3/4}.$$

The exact resulting decomposition is

$$\Delta_{2h}Y(n) = A_h(n) + \tfrac{3}{2}\beta(n+2h)^{3/4} - \tfrac{3}{2}\theta(n)\Delta_{2h}(n^{3/4}) + \Delta_{2h}E(n). \tag{B.10}$$

The θ term and E term cost, over the full correlation,

$$O(|t|hP^{3/4} + |t|P^{1/4}) = O_C(P^{7/8}). \tag{B.11}$$

Taylor expansion of the smooth A_h, with derivatives, gives

$$A_h'' = O(h^2P^{-7/4}).$$

On each branch β =3h $\sqrt{x}$ + $O(1 + h^2P^{-1/2})$ as a comparison of values. Holding β fixed in differentiation, the main retained phase therefore has

$$\Big(tA_h + \tfrac{3t}{2}\beta(x+2h)^{3/4}\Big)'' = -\tfrac{27}{32}thx^{-3/4}(1+o(1)). \tag{B.12}$$

The o(1) is uniform for 1 ≤ h ≤ H and for all admissible branches.

Write $M_h = |t|hP^{-3/4}$. The ratio of the combined D1 curvature in (B.7) to M_h is

$$O_{C,M}\big((|t|h)^{-1} + P^{-1/2}/|t|\big).$$

Choose a sufficiently large constant $h_0 = h_0$ (C,M), independent of P. For h ≥ h_0 this ratio is smaller than a fixed fraction of the lower constant in (B.12). The twist ratio in (B.9) is $O_C(P^{-1/3}/|t|)$. Thus all these terms preserve the one-signed main curvature. This argument permits arbitrary signs and mutual cancellation among the D1 coefficients.

There are only O_{C,M}(1) shifts h< h_0. The trivial bound | V_h | ≤ C P for these shifts contributes O_{C,M}(P^2/H) to (B.8). Small values of P for which H< h_0 are covered by the implied constant.

B.3 Insert the D2 reduction on its original runs

Use (A.9) of Appendix A.1 at this h, with Q=floor($P^{5/16}$). Translations of the base and coefficient by $O_C(P^{1/24})$ preserve all derivative comparisons and run counts in that proof. Conjugation handles σ =-1, and σ =0 contributes the constant factor one.

For each factor, outside an error whose sum of absolute values is $O_C(P^{15/16}\log P)$, the retained modes are

$$e(\Phi_{\ell,q}), \qquad \Phi_{\ell,q} = (N_\ell + q - B_\ell)G_\ell, \quad B_\ell = \Delta_{2h}c_\ell, \quad N_\ell = \lfloor B_\ell \rfloor,$$

or their conjugates. Here |q| ≤ Q and on each coefficient window the total coefficient sup norms plus variations are $O(\log P)$. For a fixed number of factors these become a fixed power of log P.

Every such mode has

$$|\Phi''_{\ell,q}| \ll_C P^{-15/16}. \tag{B.13}$$

The ratio to M_h is $O_C(P^{-3/16}/(|t|h))$, and so preserves (B.12). The additional coefficient-window count is

$$O_C(1 + k_\ell h P^{1/8}) = O_C(P^{1/4});$$

the original D2 run count is $O_C(P^{13/24})$. Both are retained in the common partition below.

For clarity, the $P^{15/16}$ error includes floor mismatches and floor-crossing events. Those are compared in absolute value with unimodular expressions and then replaced on full intervals. Their counts are taken on the original D2 runs before any extra refinement. Allowing fresh D2 labels on every added input cell would not be justified by that count and is explicitly excluded in Theorem B.1.

B.4 Reassemble the actual first-level carries

Apply Lemma A.2 to the finitely many first-floor evaluations in the main wave and first-difference terms. Their even shifts are $O(h + P^{1/24})$. Intersect the resulting gap runs with the stable input cells and the coefficient windows from Lemma A.1. The full number of cells is

$$D_h \ll_{C,M} (h + P^{1/24})P^{1/2}. \tag{B.14}$$

The input partition contributes $O(P^{13/24})$ cells; each floor-factor coefficient contributes $O(P^{1/4})$ additional windows. Endpoint-order cuts multiply this number by a fixed constant.

At carry cutoff R=floor($P^{1/4}$), Lemma A.2 supplies the zero coefficients with bounded variation, nonzero endpoint modes with weights $O(1/|v|)$ and phases vX(x+s), and total positive error $O(P^{5/6}\log P)$. This error is charged once over the common input interval, multiplied only by the logarithmic coefficient masses from the earlier expansions.

B.5 Sum every retained mode and every cell

After the preceding replacements, the zero carry mode has one-signed curvature comparable to M_h, uniformly in the D2 indices. On a cell of length L the second-derivative test gives $O(L\sqrt{M_h}+M_h^{-1/2})$, also for any subinterval and the odd lattice. Partial summation applies to η and all Fourier coefficients. Their total sup norms plus variations on each cell are at most a fixed power of log P. Summing lengths and all D_h endpoints gives

$$\begin{aligned} P\sqrt{M_h}+D_hM_h^{-1/2} \ll_{C,M} &(|t|h)^{1/2}P^{5/8} \\ &+\left(\sqrt{h/|t|}+P^{1/24}(|t|h)^{-1/2}\right)P^{7/8}. \end{aligned} \tag{B.16}$$

For $h_0 \le$ h $\le P^{1/12}$ and the stated t range this is $O_{C,M}(P^{11/12})$, before logarithmic losses.

For a nonzero carry Fourier mode v, its extra curvature is vX' '(x+s), of size |v| $P^{-1/2}$. All other curvatures together are $O_{C,M}(M_h)$, and

$$\frac{M_h}{|v|P^{-1/2}} \ll_{C,M} P^{-1/8}.$$

Thus these modes also have one-signed curvature of the claimed size. Summing their 1/|v| weights and all cell endpoints gives

$$R^{1/2}P^{3/4}+D_hP^{1/4} \ll_{C,M} P^{7/8}+(h+P^{1/24})P^{3/4}. \tag{B.17}$$

This is $O_{C,M}(P^{7/8})$ in the full range. A common reference curvature and uniform cell count make this bound valid after summing the coefficient masses of every D2 mode; no factor Q or D_h is hidden in that mass.

Combining the absolute errors (B.6), (B.11), (A.12), the D2 error, the coarse-boundary cost, and (B.16)–(B.17), for every h $\ge h_0$,

$$|V_h| \ll_{C,M} P^{15/16}(\log(2P))^B, \tag{B.18}$$

where B depends only on M. The largest cost is the D2 error. Substitution in (B.8), with the finitely many small shifts treated as in Appendix B.2, yields

$$|U|^2 \ll_{C,M} P^{23/12}+P^{31/16}(\log(2P))^B.$$

Since 23/12<31/16, this proves (B.3), absorbing logarithms into an arbitrarily small positive power of P.

Appendix C. Proof of the OOOEE mixed-mode theorem

Put

$$\begin{gathered} X(n)=n^{3/2}, \quad m(n)=\lfloor X(n)\rfloor, \quad Y(n)=m(n)^{3/2}, \\ v(n)=\lfloor Y(n)\rfloor, \quad Z(n)=v(n)^{3/2}, \\ w(n)=\lfloor Z(n)\rfloor, \quad U(n)=\sqrt{w(n)}. \end{gathered}$$

For every fixed $C \ge 1$ and every $\varepsilon > 0$, uniformly over nonzero integer quadruples with $\max(|i|,|j|,|k|,|l|) \le CP^{1/24}$,

$$\left|\sum_{\substack{P<n\le 2P \\ n \text{ odd}}} e\left(\frac{iX(n)+jY(n)+kZ(n)+lU(n)}{2}\right)\right| \ll_{C,\varepsilon} P^{127/128+\varepsilon}. \tag{C.1}$$

Here $e(x) = \exp(2\pi i x)$. These coordinates form a formal chain for every odd n; they coincide with the Juggler trajectory on the specified OOOEE sign class.

The counting consequences are

$$\#\{n \le N : \mathrm{word}_5(n) = OOOEE\} = \frac{N}{32} + O_\varepsilon(N^{127/128+\varepsilon}), \tag{C.2}$$

and the class $\mathcal{C}_5$ of starts with a power-envelope contraction certificate of length at most five satisfies

$$\#(\mathcal{C}_5 \cap [1, N]) = \frac{7N}{8} + O_\varepsilon(N^{127/128+\varepsilon}). \tag{C.3}$$

This is a certificate-class density, not the exact density of all starts that happen to descend within five steps.

C.1 The full phase

The proof uses Lemmas A.1–A.4 and Theorem B.1. The elementary small-shift argument is Lemma 4.4. The fractional-floor exception counts needed below are derived from (A.3)–(A.4) on original runs. The exact centered master identity supplies the signed inventory.

The bare kernel bound alone is insufficient. Writing $\theta = \{X\}$ and $\vartheta = \{Y\}$, Taylor's theorem gives

$$\begin{aligned} Z &= m^{9/4} - \tfrac{3}{2} m^{3/4}\vartheta + O(P^{-9/8}), \\ m^{3/4} &= n^{9/8} + O(P^{-3/8}), \\ U &= m^{9/8} + O(P^{-9/16}). \end{aligned} \tag{C.4}$$

For the last assertion, $\sqrt{\lfloor Z \rfloor} = v^{3/4} + O(P^{-27/16})$ and $v^{3/4} = Y^{3/4} + O(P^{-9/16})$. All comparisons are uniform for n in [P,3P].

Let $c(x) = 3kx^{9/8}/4$. Replacing the original phase by

$$F(n) = \tfrac{i}{2} X(n) + \tfrac{j}{2} Y(n) + \tfrac{k}{2} m(n)^{9/4} + \tfrac{l}{2} m(n)^{9/8} - c(n)\vartheta(n) \tag{C.5}$$

costs in the whole exponential sum at most

$$O_C\big(|k|P^{5/8} + |k|P^{-1/8} + |l|P^{7/16}\big) = O_C(P^{2/3}). \tag{C.6}$$

We keep both powers of m in (C.5) until after differencing. In particular, the fifth-coordinate term is not expanded initially into a Fourier series with a growing coefficient.

C.2 The cases k=0, including every zero coordinate

In these cases (C.4) leaves the phase $iX/2 + jY/2 + lm^{9/8}/2$, with error already covered by (C.6).

If j is nonzero, apply one A-process with $H = \lfloor P^{1/12} \rfloor$. On a first-floor gap cell and carry branch let $g = m(n + 2h) - m(n)$, with g fixed and comparable to $h\sqrt{P}$. For

$$V_g(z) = (z + g)^{9/8} - z^{9/8}$$

the exact integral formula gives

$$|V_g'(X)| \ll hP^{-13/16}, \qquad |(V_g \circ X)''| \ll hP^{-21/16}. \tag{C.7}$$

Replace $lV_g(m)/2$ by $lV_g(X)/2$ at the global cost $O(|l|hP^{3/16}) \le O_C(P^{5/16})$. In the proof of Lemma 4.4, add this last smooth term separately to each of the two g branches. No new gap cell or carry is needed. Its curvature relative to the main one $|j|hP^{-3/4}$ is

$$O(|l/j|P^{-9/16}) = O_C(P^{-25/48}).$$

The exact carry interpolation (4.8), global positive Fourier errors, and domination of the nonzero first-floor Fourier modes therefore remain valid. Both signs of j are handled by conjugating the whole phase. The correlation is $O_C(P^{7/8}(1+\sqrt{h}))$; the A-process gives $O_C(P^{23/24})$.

If j=0 and l is nonzero, expand once:

$$\tfrac{l}{2}m^{9/8} = \tfrac{l}{2}n^{27/16} - B(n)\theta + O(|l|P^{-21/16}), \quad B(x) = \tfrac{9l}{16}x^{3/16}. \tag{C.8}$$

Set $N_B = \lfloor B \rfloor$ and expand the bounded residual $B - N_B$ at cutoff $T = \lfloor P^{1/8} \rfloor$. The retained phases are

$$f_r(x) = \tfrac{l}{2}x^{27/16} + (i/2 + r - N_B)x^{3/2}, \qquad |r| \le T.$$

Freeze N_B when differentiating. Substituting its value only after differentiation gives

$$f_r'' = \tfrac{81l}{512}x^{-5/16} + O((|i| + T + 1)P^{-1/2}). \tag{C.9}$$

The coefficient is $(1/2)(27/16)(11/16) - (9/16)(3/4) = 81/512$. The error relative to the leading term is $O_C(P^{-7/48} + P^{-1/16})$. There are $O(1 + |l|P^{3/16})$ center windows; the Fourier coefficient mass and variation on each are $O(\log P)$. The summed second-derivative bound is

$$O\left(|l|^{1/2}P^{27/32} + (1 + |l|P^{3/16})|l|^{-1/2}P^{5/32}\right)\log^C(2P) \ll_C P^{83/96}\log^C(2P).$$

The positive Fourier error totals $O(P^{7/8}\log P)$. For example (A.12)'s distance-band proof gives $O(P\log(2T)/T + P^{5/6}\log(2P))$ at this cutoff. Thus this case is $O_C(P^{7/8}\log^C P)$.

Finally j=l=0 forces i nonzero. The smooth phase $iX/2$ has curvature comparable to $|i|P^{-1/2}$ and sum $O_C(P^{37/48} + P^{1/4})$. This covers every k=0 mode without using (C.1).

C.3 The twice-differenced inventory when k is nonzero

Conjugate all four frequencies if necessary to assume k ≥ 1. Use the ranges in Appendix A.3

$$H_1 = \lfloor P^{1/48} \rfloor, \quad H_2 = \lfloor P^{1/24} \rfloor, \quad 1 \le h_a < H_a, \quad d_a = 2h_a, \quad p = h_1h_2, \quad \Pi = kp \ll_C P^{5/48}. \tag{C.10}$$

Write $\Delta_a f(x) = f(x + d_a) - f(x)$, $W_a = \Delta_a Y$, $D = \Delta_1\Delta_2 Y$, and $c_{11}(x) = c(x + d_1 + d_2)$. The double difference of (C.5) is exactly

$$\tfrac{k}{2}\Delta_1\Delta_2(m^{9/4}) + \tfrac{l}{2}\Delta_1\Delta_2(m^{9/8}) + \tfrac{i}{2}\Delta_1\Delta_2 X + \tfrac{j}{2}D - \Delta_1\Delta_2(c\vartheta). \tag{C.11}$$

Apply the conjugate of the master expansion (A.15)–(A.18) to the last term, with $J = \lfloor P^{1/24} \rfloor$. Set

$$B_2 = \Delta_2 c(x + d_1), \quad B_1 = \Delta_1 c(x + d_2), \quad N_a = \lfloor B_a \rfloor.$$

The retained modes have phase

$$\begin{aligned} &\tfrac{k}{2}\Delta_1\Delta_2(m^{9/4}) + \tfrac{l}{2}\Delta_1\Delta_2(m^{9/8}) - c_{11}\{D\} \\ &\quad + tY + a_1W_1 + a_2W_2 + qD + \phi_0, \qquad a_1 = -N_2 + u, \quad a_2 = -N_1 + v. \end{aligned} \tag{C.12}$$

Here t,u,v are integers of absolute value $O_C(J)$; $q \in \frac{1}{2}\mathbb{Z}$ of absolute value $O_C(J)$, including the original j/2. The function ϕ_0 is $i\Delta_1\Delta_2X/2$ plus a bounded integer combination of B_1, B_2, c_{11}. In particular

$$\begin{aligned} |a_1|h_1 + |a_2|h_2 &\ll_C \Pi P^{1/8} + J(h_1 + h_2) \ll_C P^{11/48}, \\ |\phi_0''| &\ll_C kP^{-7/8} + |i|pP^{-5/2}. \end{aligned} \tag{C.13}$$

All signs of i,j,l,u,v,q are permitted. The original jY/2 contributes only jD/2 after two differences, so t remains an integer.

The full coefficient mass is $O(\log^7(2P))$. Each multi-index has a fixed harmonic envelope under which its normalized coefficient has bounded sup norm plus variation on the N_1, N_2 cells. The centers have local density $O_C(P^{-19/24})$. These are exactly the centering and five binary-carry operations in the master-inventory argument, with signs reversed. The deleted first master term costs $O_C(P^{11/48})$, and all positive errors cost $O_C(P^{23/24}\log^C P)$. Their arguments are the same Y corners, W_1, W_2,the shifted W_1, and D. The new factors in (C.11) have modulus one, so the positive bounds (A.13) are unchanged. In particular no OOOEE discrepancy has been assumed to control them.

C.4 Frozen branches and the extra powers

Partition first into the original gap runs determined by the floors of $\Delta_1 X, \Delta_2 X, \Delta_{d_1+d_2} X$. Their total count is $O_C(P^{13/24})$, their local count in length L is $O_C(1 + LP^{-11/24})$, and their lengths are $O_C(\sqrt{P})$. On a carry pattern let

$$m(n + d_a) = m(n) + \beta_a, \quad m(n + d_1 + d_2) = m(n) + \beta_1 + \beta_2 + b.$$

Then $-1 \le b \le 2$, since the real mixed difference of X is positive and o(1). The β values and b are fixed on the branch; $\beta_a = 3h_a\sqrt{x} + O(1 + h_a^2 P^{-1/2})$ only as values.

For any exponent a define

$$F_a(z) = (z + \beta_1 + \beta_2 + b)^a - (z + \beta_1)^a - (z + \beta_2)^a + z^a.$$

The formulas used for every derivative are

$$\begin{aligned} F_{a,0}(z) &= a(a-1)\int_0^{\beta_1}\int_0^{\beta_2}(z+s+t)^{a-2}\,dt\,ds, \\ F_a(z) - F_{a,0}(z) &= ab\int_0^1 (z + \beta_1 + \beta_2 + \tau b)^{a-1}\,d\tau. \end{aligned} \tag{C.14}$$

They apply also when b<0. The notation $F_{a,0}$ means b=0. Put $G = F_{3/2}(X)$ and $A = \lfloor F_{3/2}(m) \rfloor$. The exact identity $\Delta_1\Delta_2(m^a) = F_a(m)$ holds on its actual carry pattern.

The fifth-coordinate term can now be replaced directly by $lF_{9/8}(X)/2$. The relevant bounds are

$$\begin{aligned} |F'_{9/8}(X)| &\ll |b|P^{-21/16} + pP^{-29/16}, \\ |(F_{9/8} \circ X)''| &\ll |b|P^{-29/16} + pP^{-37/16}, \\ |(F_{9/8} \circ X)'''| &\ll |b|P^{-45/16} + pP^{-53/16}. \end{aligned} \tag{C.15}$$

The total replacement error is $O(|l|P^{-5/16} + |l|pP^{-13/16}) = O_C(1)$. Replacing qD by qG costs $O_C(P^{7/24})$, by (A.4) and the mean value theorem; integrality of q is irrelevant for that comparison.

Combine the two large terms before expanding in θ:

$$H(z,x) = \tfrac{k}{2}F_{9/4}(z) - c_{11}(x)F_{3/2}(z), \qquad B_{\text{core}}(x) = \partial_z H(X(x), x).$$

Since $-c_{11}\{D\} = -c_{11}F_{3/2}(m) + c_{11}A$, the core of (C.12) is H(m,x)+ c_{11} A. Taylor expansion gives

$$H(m,x) = H(X,x) - B_{\text{core}}\theta + O_C(k|b|P^{-9/8} + \Pi P^{-13/8}), \tag{C.16}$$

so its total error is bounded. The growing coefficient is

$$B_{\text{core}} = \tfrac{27}{32}kbx^{3/8} + O_C\big(\Pi P^{-1/8} + k(h_1 + h_2)P^{-5/8}\big). \tag{C.17}$$

The leading coefficient is $45/32 - 9/16 = 27/32$. The remainder after subtracting $(27/32)kbx^{3/8}$ is bounded in (C.10). Its derivative on each original run is $O_C(P^{-3/4})$, by (C.14) with the offsets fixed. Each such run has length $O_C(P^{1/2})$, so this remainder has bounded variation there. The leading term itself has derivative $O_C(kP^{-5/8})$; it is not included in the smaller remainder-derivative bound.

All carry patterns are retained. As in (A.12), their indicators are finite unions of arcs in θ. Endpoint-order cuts add only a fixed multiple of the run count. At cutoff $R_c = P^{1/4}$, each nonzero endpoint mode has weight $O(1/|s|)$ and phase sX(x+d), for an allowed shift d; the zero coefficient has bounded variation. Their global positive errors are $O(P^{5/6}\log P)$. We use $R = P^{5/16}$ for the bounded θ coefficients below; their global errors have the same bound. No error is charged again on each gap run or frequency window.

C.5 Nonzero total Y frequency

Suppose t is nonzero in (C.12). Keep the first differences $a_1W_1 + a_2W_2$ unexpanded and keep the floor A on its original fixed-label runs. In particular do not cut into A's integer levels.

Center the growing term (C.17) at

$$N_* = \left\lfloor \tfrac{27}{32}kbx^{3/8} \right\rfloor.$$

On the intersections of original runs and N_* windows, $B_{\mathrm{core}} - N_*$ has bounded size and variation: the leading fractional part is monotone with variation at most one, and the remainder has the variation bound just proved. This residual need not lie in $[0,1]$. For $b = 0$ use $N_* = 0$. Expand it at R using the bounded-residual extension of Lemma 4.7. After the first-floor carry expansion the smooth part is

$$\Phi = H(X,x) + \tfrac{l}{2}F_{9/8}(X) + qG + \phi_0 + (r - N_*)X + sX(x+d).$$

Its third derivative satisfies

$$|\Phi'''| \ll_C kP^{-9/8} + \Pi P^{-13/8} + (R + |N_*| + R_c)P^{-3/2} + |l|(P^{-45/16} + pP^{-53/16}) \ll_C P^{-13/12}. \quad \text{(C.18)}$$

The q and ϕ_0 terms are smaller; $|N_*| \ll_C P^{5/12}$.

The common coarse partition consists of original runs, N_1, N_2 windows, N_* windows, and endpoint-order cuts. Its local count remains $O_C(1 + LP^{-11/24})$, since N_* has local density $O_C(kP^{-5/8}) = O_C(P^{-7/12})$. Fixed b on each original run permits this assertion even though b changes between runs. The assigned D2 labels do not change at extra refinements. An empty carry pattern is extended using an admissible fixed label on that original run and zero amplitude there.

Every normalized retained term now satisfies (B.3): the nonzero integer t is in its allowed range; (C.13) fits its widened first-difference budget; (C.18) fits its twist budget; and the single floor term is + c_{11} A, namely σ =-1 in (B.3). The sum of all these weighted terms is

$$O_{C,\varepsilon}(P^{31/32+\varepsilon}). \quad \text{(C.19)}$$

The D2 reduction is invoked inside (B.3), after that theorem's additional A-process, rather than on an arbitrary undifferenced floor.

C.6 Zero total Y frequency: shared reductions

Now t=0. Replace A by $J_F = \lfloor G \rfloor$. For b nonzero, (A.3)–(A.4), at derivative scale $P^{-1/4}$ and distance $O(P^{-3/4} + pP^{-5/4})$, give $O_C(P^{3/4})$ exceptions on all original runs. For b=0 the scale is $pP^{-3/4}$, the distance is $O(pP^{-5/4})$, and the same count is $O_C(pP^{1/4} + (h_1 + h_2)P^{1/2}) =$

$O_C(P^{13/24})$. Each changed exponential costs at most two. The coefficient c_{11} is not multiplied by the size of a floor error.

For $V_a(z) = (z+\beta_a)^{3/2} - z^{3/2}$, expand

$$a_aV_a(m) = a_aV_a(X) - a_aV_a'(X)\theta + O(|a_a|h_aP^{-7/4}).$$

By (C.13) the global error is $O_C(P^{-25/48})$. The smooth part and θ coefficient are consequently

$$\begin{aligned} F_{\mathrm{sm}} &= \tfrac{k}{2}F_{9/4}(X) - c_{11}(G-J_F) + a_1V_1(X) + a_2V_2(X) + \tfrac{l}{2}F_{9/8}(X) + qG + \phi_0, \\ B &= B_{\mathrm{core}} + a_1V_1'(X) + a_2V_2'(X). \end{aligned} \tag{C.20}$$

The wave contribution to B is bounded:

$$\left|\sum a_aV_a'(X)\right| \ll_C \Pi P^{-1/8} + J(h_1+h_2)P^{-1/4} \ll_C 1.$$

Its derivative on a frozen cell is bounded by $O_C(\Pi P^{-9/8} + J(h_1+h_2)P^{-5/4})$. Multiplying the derivative bound by the original-run length $O_C(P^{1/2})$ proves bounded variation on every frozen cell; refining a cell cannot increase this bound. Together with the remainder bound following (C.17), this verifies the hypotheses of the bounded-residual extension of Lemma 4.7 in both offset cases. All integer levels of J_F are now included in the derivative-test partition.

C.7 Nonzero b: recompute the mixed curvature

For b nonzero, center B at the same N_* as in Appendix C.5; its bounded residual has bounded variation. Retained phases are $f = F_{\mathrm{sm}} + (r-N_*)X + sX(x+d)$.

Freeze $\beta_a, b, J_F, N_*, N_a$ before differentiation. The three leading contributions, in units of $kbx^{-1/8}$, are

term	coefficient
$(kF_{9/4}(X)/2)''$	$945/512$
$(-c_{11}(G-J_F))''$	$-864/512$
$-N_*X''$	$-324/512$

Their sum is

$$-\tfrac{243}{512}. \tag{C.21}$$

For the middle row use $-2c_{11}'G' - c_{11}G'' - c_{11}''(G-J_F)$. The last factor $G-J_F$ is bounded, so its curvature term is only $O(kP^{-7/8})$. Dropping the integer-floor contribution would give the wrong leading coefficient.

Uniformly for every retained index,

$$f'' = -\tfrac{243}{512}kbx^{-1/8} + O_C(P^{-3/16}). \tag{C.22}$$

For clarity, the error terms include

$$\begin{aligned} &\Pi P^{-5/8} + k(h_1+h_2)P^{-9/8} + kP^{-7/8} \\ &+ (|a_1|h_1 + |a_2|h_2)P^{-3/4} + (R+R_c+1)P^{-1/2} + R_c(h_1+h_2)P^{-3/2} \\ &+ |q|(P^{-5/4} + pP^{-7/4}) + |l|(P^{-29/16} + pP^{-37/16}) + |\phi_0''|. \end{aligned}$$

Every term fits (C.22) under (C.10),(C.13). Both signs of b are allowed. The error is smaller than the leading curvature by $O_C(P^{-1/16}/k)$.

The full number of intervals is $O_C(P^{3/4})$. The G-levels add at most $O_C(P^{3/4} + P^{13/24})$; the original gaps, moving centers, and arc-order cuts are smaller. Summing lengths and every endpoint in the second-derivative estimate, with coefficient masses and variation, gives

$$O_C\left(\sqrt{k}P^{15/16} + k^{-1/2}P^{13/16}\right)\log^C(2P) \ll_C P^{23/24}\log^C(2P). \tag{C.23}$$

The positive errors and floor exceptions are smaller. This proves the nonzero-offset part for the actual mixed family.

C.8 Zero b: the actual negative moving centers

When b=0, (C.20) has $|B| \ll_C 1$ and bounded variation on every original run intersected with the N_a windows. Expand it directly at R, without a new θ center. After the carry expansion put $\ell_F = r + s$, an integer, and $\alpha_0 = uh_1 + vh_2$. For fixed ℓ_F, the coefficient sup norms plus variations sum to $O(\log(2P)/(1+|\ell_F|))$. Indeed the convolution of the two coefficient envelopes satisfies

$$\sum_{r+s=\ell_F} \frac{1}{(1+|r|)(1+|s|)} \ll \frac{\log(2P)}{1+|\ell_F|}.$$

Split at $|r| \geq |\ \ell_F\ |/2$ or $|s| \geq |\ \ell_F\ |/2$ to verify this bound. The same estimate holds for the coefficient variations by the product variation inequality; the phases for different r,s need not be identical.

The pure $kF_{9/4}(X)/2$ curvature has leading coefficient $-6075/2048$ in units of $\Pi x^{-5/8}$. The negative fractional anchor contributes $243/128 = 3888/2048$, so their subtotal is $-2187/2048$. The centers in this problem have negative signs:

$$h_1a_1 + h_2a_2 = -\tfrac{27}{8}\Pi x^{1/8} + \alpha_0 + O_C(h_1 + h_2 + \Pi(h_1 + h_2)P^{-7/8}). \tag{C.24}$$

The frozen wave curvature is $-27(h_1a_1+h_2a_2)x^{-3/4}/32$, up to smaller terms. It adds $729/256 = 5832/2048$ to the subtotal. Thus

$$f'' = -\tfrac{27}{32}\alpha_0 x^{-3/4} + \tfrac{3645}{2048}\Pi x^{-5/8} + \tfrac{3}{4}\ell_F x^{-1/2} + O_C(\Pi P^{-3/4}). \tag{C.25}$$

This is the coefficient for the complete mixed phase (C.12), including the pure power and both moving centers.

The errors in (C.25) include

$$\begin{aligned}&(h_1 + h_2)P^{-3/4} + (|a_1| + |a_2|)P^{-5/4} + (|a_1|h_1^2 + |a_2|h_2^2)P^{-7/4}\\ &+ k(h_1 + h_2)P^{-9/8} + \Pi(h_1 + h_2)P^{-13/8} + kP^{-7/8} + |q|pP^{-7/4}\\ &+ P^{-29/24} + |l|pP^{-37/16} + |\phi_0''|.\end{aligned}$$

Use $h_1 + h_2 \leq 2p \leq 2\Pi$, before inserting independent shift caps, for the leading rounding error. All terms have the claimed bound. The fifth-coordinate term is particularly small here.

There is a strict frequency gap:

$$\frac{|\alpha_0|P^{-3/4}}{\Pi P^{-5/8}} \ll_C JP^{-1/8} = O_C(P^{-1/12}), \qquad \frac{\Pi P^{-5/8}}{P^{-1/2}} \ll_C P^{-1/48}. \tag{C.26}$$

Consequently $\ell_F = 0$ has one-signed curvature comparable to $\Pi P^{-5/8}$; every nonzero integer ℓ_F has one-signed curvature comparable to $|\ell_F|P^{-1/2}$. No hypothesis on the largest individual signed wave is used.

The common partition has $O_C(P^{13/24})$ cells, including $O_C(pP^{1/4}+P^{13/24})$ zero-offset G-level cuts. For $\ell_F = 0$, summing all lengths and endpoints costs

$$O_C\left(\Pi^{1/2}P^{11/16} + \Pi^{-1/2}P^{41/48}\right)\log^C(2P).$$

For nonzero ℓ_F, sum the harmonic weights as well:

$$O_C\left(R^{1/2}P^{3/4} + P^{13/24+1/4}\right)\log^C(2P) = O_C(P^{29/32}\log^C(2P)). \tag{C.27}$$

The zero mode and positive errors are smaller. This directly estimates the moving coefficients in (C.12), with every window endpoint included.

C.9 Assembly and the dyadic count

For the double correlation of (C.5), the accounting is:

Contribution	Exponent, apart from fixed logarithmic powers
Deleted master term	11/48
Positive short-Fourier errors	23/24
Extra-power Taylor errors and slow D replacement	at most 7/24
Nonzero t, via (B.3)	31/32
t=0, nonzero b	23/24
t=0, b=0	29/32

The first-floor Fourier errors and floor mismatches are included in the last three estimates. Thus the double correlation is $O_{C,\varepsilon}(P^{31/32+\varepsilon})$. For the sum S_F and its first and second correlations, van der Corput differencing on the odd lattice gives

$$|S_F|^2 \ll P^2/H_1 + (P/H_1) \sum_{h_1<H_1} |T_1(h_1)|,$$

$$|T_1(h_1)|^2 \ll P^2/H_2 + (P/H_2) \sum_{h_2<H_2} |T_2(h_1,h_2)|.$$

Substitution gives

$$|T_1|^2 \ll_C P^{47/24} + P^{63/32+\varepsilon}, \qquad |T_1| \ll_{C,\varepsilon} P^{63/64+\varepsilon},$$

and

$$|S_F|^2 \ll_C P^{95/48} + P^{127/64+\varepsilon}, \qquad |S_F| \ll_{C,\varepsilon} P^{127/128+\varepsilon}.$$

Overlap endpoints cost $O(h_1+h_2)$ and fit these estimates. Arbitrarily small epsilon losses may be relabeled after each step. Together with Appendices C.1–C.2 this proves (C.1).

Apply the four-dimensional Erdős–Turán–Koksma inequality to

$$(\{X/2\},\{Y/2\},\{Z/2\},\{U/2\})$$

on the odd integers in a single dyadic block. Use cutoff $\lfloor P^{1/24} \rfloor$. Its diagonal error is $O(P^{23/24})$; the weighted nonzero modes in (C.1) cost $O_\varepsilon(P^{127/128+\varepsilon})$, absorbing the four harmonic sums. All sixteen half-boxes have volume 1/16. Since there are P/2+ $O(1)$ odd inputs, each formal sign class has count

$$P/32 + O_\varepsilon(P^{127/128+\varepsilon})$$

on this block. A half-box records exactly the parity of the floor: even for a coordinate in [0,1/2), odd for one in [1/2,1).

Use this discrepancy argument separately on each dyadic block in a decomposition of [1,N], with that block's own frequency cutoff. Then sum the counts and errors. One must not impose the largest block's cutoff on all smaller blocks. Taking a small epsilon first, the error series is geometric; the remaining bounded initial interval contributes $O(1)$. The class with m,v odd and w,floor(U) even is exactly OOOEE for odd n. This proves (C.2), including endpoint conventions. All fifteen nonempty products of its four parity signs also have this error bound, by summing the sixteen sign classes. In the manuscript's notation, H(OOOEE; δ) follows for every $0 < \delta <$ 1/128.

Finally Lemma 5.1 and Theorem 5.3 of the manuscript give the disjoint union of its 27/32 class and OOOEE. The former error $O(N^{47/48})$ is smaller. Their densities add to $27/32+1/32 =$ 7/8, proving (C.3). This proves Corollary 4.12 and Theorem 5.4.

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